

MATH406 Notes

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1 Intro

Number theory is mostly about the properties of integers

We will preemptively assume the entirety of set theory and the basics of lin alg.

Textbook: Divisibility

2 Lecture 1

2.1 2.1 - Divisibility

We can add/subtract/multiply integers to get integers (eg prop of field, closed).

Definition: Let a, b be integers, $b \neq 0$. We say that b divides a (denoted $b|a$) if there exists an integer k such that $a = bk$.

Additionally,

- If b does not divide a , we write $b \nmid a$.
- When $d|a$, we say that a is a multiple of d and that d is a divisor (or factor) of a .
- For any nonzero integer d we have $d|0$ since $0 = d \cdot 0$.
- We never consider division by zero, so $0|a$ is not defined.

Convention: When we talk about the set of divisors of a positive integer, we mean the set of positive divisors. For instance, the divisors of 6 are 1,2,3,6 (not -1,-2,-3,-6).

Proposition: Let a, b, c integers. Then $a|b$ and $b|c$ implies $a|c$.

Proof: Since $a|b$, there exists an integer k such that $b = ak$. Since $b|c$, there exists an integer l such that $c = bl$. Thus,

$$\begin{aligned} c &= bl \\ &= (ak)l \\ &= a(kl) \end{aligned}$$

Since kl is an integer under field closure, we have $a|c$. □

Definition: A linear combination of integers is an expression of the form $ax + by$ where x, y are integers.

"A divisor of both a and b is a divisor of any linear combination of a and b ."

Proposition: Bilinearity of division; that is, $d|ax + by$ if $d|a$ and $d|b$.

Proof: Since $d|a$, there exists an integer k such that $a = dk$. Since $d|b$, there exists an integer l such that $b = dl$. Thus,

$$\begin{aligned} ax + by &= (dk)x + (dl)y \\ &= d(kx) + d(ly) \\ &= d(kx + ly) \end{aligned}$$

Since $kx + ly$ is an integer under field closure, we have $d|ax + by$. □

Corollary: Suppose a, b, d are integers. Then, if $d|a$ and $d|b$ then $d|(a - b)$ and $d|(a + b)$.

Proof: Let $x = 1$ and $y = -1$ in the previous proposition to get $d|(a - b)$. Let $x = 1$ and $y = 1$ to get $d|(a + b)$. □

2.2 2.2 - Euclid's Theorem

Definition: A prime number is an integer $p \geq 2$ whose only positive divisors are 1 and p itself. An integer $n \geq 2$ that is not prime is called composite.

Theorem (Euclid's Theorem): There are infinitely many prime numbers.

3 Lecture 2

Proof of Euclid's Theorem: Suppose there are finitely many primes $p_1, p_2, \dots, p_n$. Let

$$Q = p_1 p_2 p_3 \cdots p_n + 1$$

Then Q is either prime or composite. If Q is prime, then Q is a prime not in our list. If Q is composite, then Q has a prime divisor p . Since p divides the product $p_1 p_2 \cdots p_n$, it follows that p does not divide $Q - p_1 p_2 \cdots p_n = 1$. Thus, p is not in our list of all primes. In either case, we have a contradiction. Therefore, there are infinitely many primes. □

3.1 2.4 - The Sieve of Eratosthenes

Theorem (Sieve of Eratosthenes): To find all primes less than or equal to a given integer $n \geq 2$, we can use the following algorithm:

1. Write down all integers from 2 to n .
2. Start with the first number p in the list (which is 2). Circle p and cross out all multiples of p greater than p .
3. Find the next number in the list that is not crossed out. If there is no such number, stop. Otherwise, let p be this new number and repeat step 2.
4. When the algorithm stops, the circled numbers are all the primes less than or equal to n .

Proof: Let p be any prime less than or equal to n . When we reach step 2 with this value of p , we circle it since it has not been crossed out (as it is prime). Thus, all primes less than or equal to n are circled when the algorithm stops. □

4 Lecture 3

4.0.1 2.5 - The Division Algorithm

Consider the grade school long division problem. We want to do, for instance, 95 divided by 7. We can write

$$95 = 7 \cdot 13 + 4$$

Theorem (The Division Algorithm): Let $a, b \in \mathbb{Z}$ and $b > 0$. Then $\exists! q, r \in \mathbb{Z}$ st

$$a = bq + r \text{ with } 0 \leq r < b$$

Note that if $b > a$ then $r = a$ and $q = 0$.

Proof: (Existence) Let q be the largest integer such

$$q \leq \frac{a}{b} < q + 1$$

Then $bq \leq a < bq + b$. and so $0 \leq a - bq < b$. Let $r = a - bq$. Then $a = bq + r$ with $0 \leq r < b$.

(Uniqueness) Suppose there exist q', r' such that

$$a = bq' + r' \text{ with } 0 \leq r' < b$$

Then

$$\begin{aligned} bq + r &= bq' + r' \\ b(q - q') &= r' - r \end{aligned}$$

If $q \neq q'$, then $|b(q - q')| \geq b$ since $|q - q'| \geq 1$. But $|r' - r| < b$ since both r and r' are between 0 and b . This is a contradiction. Thus, $q = q'$, which implies $r = r'$. $\square$

Example:

1. For $a = 100, b = 9$, we have $100 = 9 \cdot 11 + 1$.
2. For $a = -100, b = 9$, we have $-100 = 9 \cdot (-12) + 8$.

Corollary: $a|b$ iff the remainder when b is divided by a is 0.

Proof: Let $b = aq + r$ by the division algorithm. If $r = 0$, then $b = aq$ and so $a|b$. Conversely, if $a|b$, then $b = ak$ for some integer k . By the uniqueness part of the division algorithm, we must have $q = k$ and $r = 0$. $\square$

4.0.2 2.6 - The GCD

Fact: If you have a nonzero integer a , then it has finite divisors by the well-ordering principle. (eg there are only finitely many positive integers less than or equal to $|a|$, and thus finite candidates for divisors).

Definition (GCD): Let a, b be integers, not both zero. The greatest common divisor of a and b , denoted $\gcd(a, b)$, is the largest positive integer that divides both a and b .

Example:

- $\gcd(12, 15) = 3$

- $\gcd(0, 5) = 5$
- $\gcd(0, 0)$ is not defined.

Definition (Relatively Prime): Two integers are relatively prime if their gcd is 1.

Example:

- $\gcd(8, 15) = 1$ so 8 and 15 are relatively prime.
- $\gcd(9, 28) = 1$ so 9 and 28 are relatively prime.
- $\gcd(0, 1) = 1$ so 0 and 1 are relatively prime.

Corollary: $\gcd(p, n) = 1 \iff p \nmid n$.

Proof: If $\gcd(p, n) = 1$, then the only positive divisor of both p and n is 1. Thus, p does not divide n . Conversely, if $p \nmid n$, then the only positive divisor of both p and n is 1, so $\gcd(p, n) = 1$. $\square$

5 Lecture 4

Last time, we defined the GCD.

Proposition (2.10): Let $a, b \in \mathbb{Z}$ and $d = \gcd(a, b)$. Then,

$$\gcd\left(\frac{a}{d}, \frac{b}{d}\right) = 1$$

Proof: Recall that the gcd is defined to be at least 1, always. We will prove that

$$\gcd\left(\frac{a}{d}, \frac{b}{d}\right) \leq 1$$

To do this, we will show that any common divisor of $\frac{a}{d}$ and $\frac{b}{d}$ is ≤ 1 .

Suppose that for some positive integer c , $c \mid \frac{a}{d}$ and $c \mid \frac{b}{d}$. By def of divisibility we thus have that $\frac{a}{d} = c \cdot k$, and that $\frac{b}{d} = c \cdot l$ for $k, l \in \mathbb{Z}$.

Then, $a = c \cdot d \cdot k$, and $b = c \cdot d \cdot l$. This shows that cd is a common divisor of a and b . However, d is the gcd of a, b , so it must be that $cd \leq d$. Under $\mathbb{Q}$ field properties there must $\exists!$ inverse to d , $\frac{1}{d}$ and by field closure we can multiply, so $c \leq 1$. $\square$

5.1 2.7 - The Euclidean Algorithm

How can we compute gcds?

One method is to factor each one into primes, take the overlapping subset, and find their product. This is dumb for large integers because factoring large integers is not computationally feasible (for traditional computers)

Lemma: Suppose $a = bq + r$. The set of common divisors of a and b is the same as the set of common divisors of b and r . It thus follows they have the same gcd.

Proof: Suppose $a = bq + r$. Then, a is a linear combination of b, r . Then, any common divisor of b, r must also divide a by a former proposition. Thus, any common divisor of b, r is by definition a divisor of b , but also a divisor of a . Thus, they are also common divisors of a, b .

Similarly, we can rewrite the equation to say that $r = a - bq$, and thus similarly all common divisors of a, b must be common divisors of r, b . So we are done. $\square$

Note: Here we can see the idea for the euclidean algorithm presented. We can reduce the gcd problem from a, b to be r, b , where r is strictly lesser than a . In fact, it's $\leq b$ by definition of division algorithm. Then, we can do this to b , then r again, etc until the numbers are very small.

Example: Let's use this idea to compute $\gcd(315, 220)$. Then,

$$315 = 1 \cdot 220 + 95 \quad 220 = 2 \cdot 95 + 30 \quad 95 = 3 \cdot 30 + 5 \quad 30 = 6 \cdot 5 + 0$$

and so applying the lemma, we get that $\gcd(315, 220) = \gcd(220, 95) = \gcd(95, 30) = \gcd(30, 5) = \gcd(5, 0) = 5$.

or the last nonzero remainder, is the gcd.

Theorem (The Euclidean Algorithm): Let $a, b \geq 0 \in \mathbb{Z}, b \neq 0$. Then, $a = bq_1 + r_1$ with $0 \leq r_1 \leq b$. Then, recursively apply the theorem with parameters b, r . The base case applies when the remainder is 0, in which case the divisor is the gcd.

In python,

```
def gcd(a, b):
    if b==0:
        return a
    else:
        return gcd(b, a%b)
```

Proof: This is just repeated application of the lemma. $\square$

5.1.1 2.7.1 - The Extended Euclidean Algorithm

Theorem (2.12): $\exists x, y \in \mathbb{Z}$ st for $a, b \in \mathbb{Z}$ not both zero, $ax + by = \gcd(a, b)$.

Note: They are not unique.

6 Lecture 5

The idea is that we can solve for the remainder. For instance, take the example:

Ex: $\gcd(1239, 735)$ results in

$$\begin{aligned} 1239 &= 1 \cdot 735 + 504 \\ 735 &= 1 \cdot 504 + 231 \\ 504 &= 2 \cdot 231 + 42 \\ 231 &= 5 \cdot 42 + 21 \\ 42 &= 2 \cdot 21 + 0 \end{aligned}$$

But this also gives

$$\begin{aligned} 504 &= 1239 - 1 \cdot 735 \\ 231 &= 735 - 1 \cdot 504 \\ 42 &= 504 - 2 \cdot 231 \\ 21 &= 231 - 5 \cdot 42 \end{aligned}$$

We can substitute these all the way up to get

$$\begin{aligned}
 21 &= 231 - 5 \cdot 42 \\
 &= 231 - 5 \cdot (504 - 2 \cdot 231) \\
 &= 11 \cdot 231 - 5 \cdot 504 \\
 &= 11 \cdot (735 - 1 \cdot 504) - 5 \cdot 504 \\
 &= 11 \cdot 735 - 16 \cdot 504 \\
 &= 11 \cdot 735 - 16 \cdot (1239 - 1 \cdot 735) \\
 &= 27 \cdot 735 - 16 \cdot 1239
 \end{aligned}$$

This gives us $x = -16$ and $y = 27$ such that $1239 \cdot (-16) + 735 \cdot 27 = 21$.

Theorem (The Extended Euclidean Algorithm): Let $a, b \in \mathbb{Z}$ not both zero. Then, we can find $x, y \in \mathbb{Z}$ such that $ax + by = \gcd(a, b)$. Do so by:

- Apply the Euclidean algorithm to find the gcd and all the remainders.
- Rewrite each remainder as a linear combination of a and b by substituting the previous remainders.
- Continue substituting until you express the gcd as a linear combination of a and b .

In python,

```
def extended_gcd(a, b):
    if b==0:
        return (a, 1, 0)
    else:
        d, x1, y1 = extended_gcd(b, a%b)
        x = y1
        y = x1 - (a//b)*y1
        return (d, x, y)
```

Proof: This is just the algorithm described above. The correctness follows from the Euclidean algorithm and properties of linear combinations. $\square$

Example: Find x, y such that $19x + 7y = 1$.

We apply the extended euclidean algorithm:

$$\begin{aligned}
 19 &= 2 \cdot 7 + 5 \\
 7 &= 1 \cdot 5 + 2 \\
 5 &= 2 \cdot 2 + 1 \\
 2 &= 2 \cdot 1 + 0
 \end{aligned}$$

Then, in reverse,

$$\begin{aligned}
 1 &= 5 - 2 \cdot 2 \\
 &= 5 - 2 \cdot (7 - 1 \cdot 5) \\
 &= 3 \cdot 5 - 2 \cdot 7 \\
 &= 3 \cdot (19 - 2 \cdot 7) - 2 \cdot 7 \\
 &= 3 \cdot 19 - 8 \cdot 7
 \end{aligned}$$

Some applications:

Proposition (2.16): Let $a, b, c \in \mathbb{Z}$ st $a \neq 0, \gcd(a, b) = 1$. Then if $a|bc$, then $a|c$.

The idea is that if a, b are coprime, then a has no common factors with b . Thus, if a divides bc , it must be that a divides c .

Proof: Since $\gcd(a, b) = 1$, there exist $x, y \in \mathbb{Z}$ such that $ax + by = 1$. Multiplying both sides by c gives $acx + bcy = c$. Since $a|acx$ and $a|bcy$ (since $a|bc$), it follows that $a|c$. $\square$

Corollary: Let $p \in \mathbb{Z}$ prime, and $a, b \in \mathbb{Z}$. Then if $p|ab$, then $p|a$ or $p|b$.

Proof: If $p|a$, we are done. Otherwise, $\gcd(p, a) = 1$ since p is prime and does not divide a . Thus, by the previous proposition, $p|b$. $\square$

Corollary (2.14): Let $a, b, d \in \mathbb{Z}$ st $d = \gcd(a, b)$. For $e \in \mathbb{Z}$ if $e|a, e|b$, then $e|d$.

Proof: Since d is a linear combination of a and b , we have that $d = ax + by$ for some integers x, y . Since $e|a$ and $e|b$, it follows that $e|ax$ and $e|by$. Thus, $e|d$. $\square$

6.1 2.9 - Fermat and Mersenne Primes

We know that there are infinitely many primes by Euclid's theorem. However, are there infinitely many primes of certain forms? And can we use this to generate primes? Useful for cryptography, etc.

Definition (Mersenne Number): The n th Mersenne number is defined as $M_n = 2^n - 1$ for $n \geq 1$.

Proposition: If $a|n$, then $M_a|M_n$. If n is composite, then M_n is composite.

Proof: Suppose $n = ab$ for integers $a, b > 1$. Then,

$$\begin{aligned} M_n &= 2^{ab} - 1 \\ &= (2^a)^b - 1^b \\ &= (2^a - 1)((2^a)^{b-1} + (2^a)^{b-2} + \dots + 1) \end{aligned}$$

Since $a, b > 1$, it follows that $2^a - 1 > 1$ and the second factor is also greater than 1. Thus, M_n is composite. $\square$

Example: $M_{10} = 2^{10} - 1 = 1023$ is composite. We know that $2|10$ so $M_2|M_{10}$. In fact, $M_2 = 3$ and $1023 = 3 \cdot 341$. Also, $5|10$ so $M_5|M_{10}$. In fact, $M_5 = 31$ and $1023 = 31 \cdot 33$.

Note: If p is prime, is M_p prime? For some, yes, i.e. 13, 17, 19. However M_{11} isn't prime for instance.

Definition (Fermat Number): The n th Fermat number is defined as $F_n = 2^{2^n} + 1$ for $n \geq 0$.

Note: He initially looked at $2^m + 1$ but found that unless m is a power of 2, $2^m + 1$ is composite. For instance, $2^6 + 1 = 65$ is composite.

Proposition: For $2^m + 1$, if m is not a power of 2, then $2^m + 1$ is composite. If $m = 2^n$, then $2^m + 1$ is prime for $n = 0, 1, 2, 3$ but composite for $n = 4$.

Proof: Suppose m is not a power of 2. Recall that if k is odd, then $x^k + 1$ is divisible by $x + 1$. Since m is

not a power of 2, then we can write $m = 2^n \cdot k$ for some odd integer $k > 1$. Then,

$$\begin{aligned} 2^m + 1 &= 2^{2^n \cdot k} + 1 \\ &= (2^{2^n})^k + 1^k \\ &= (2^{2^n} + 1) \left((2^{2^n})^{k-1} - (2^{2^n})^{k-2} + \dots - 1 \right) \end{aligned}$$

Since $k > 1$, it follows that $2^{2^n} + 1 > 1$ and the second factor is also greater than 1. Thus, $2^m + 1$ is composite. $\square$

7 Lecture 6

Proposition: If $M_p = 2^p - 1$ is prime, then p is prime. Such an M_p is called a Mersenne prime.

Note: There are only 52 known Mersenne primes! And it's not known if there are infinitely many.

It turns out that F_5, F_6, F_7 are all composite. In fact, only F_0 to F_4 are known to be prime. It is not known if there are infinitely many Fermat primes.

8 Lecture 7

Chapter 3 now, Linear Diophantine Equations.

8.1 3.1 - Linear Diophantine Equations

Take $ax + by = c$ for $a, b, c \in \mathbb{Z}$. We are interested in integer pairs (x, y) that satisfy this linear diophantine equation.

The goal will be to describe all such solutions (x, y)

Example: Find integers x, y such that $19x + 7y = 1$.

By the extended euclidean algorithm,

$$\begin{aligned} 19 &= 2 \cdot 7 + 5 \\ 7 &= 1 \cdot 5 + 2 \\ 5 &= 2 \cdot 2 + 1 \\ 2 &= 2 \cdot 1 + 0 \end{aligned}$$

Then, in reverse,

$$\begin{aligned} 1 &= 5 - 2 \cdot 2 \\ &= 5 - 2 \cdot (7 - 1 \cdot 5) \\ &= 3 \cdot 5 - 2 \cdot 7 \\ &= 3 \cdot (19 - 2 \cdot 7) - 2 \cdot 7 \\ &= 3 \cdot 19 - 8 \cdot 7 \end{aligned}$$

Then, one solution is $3, -8$. There are infinitely many solutions, since we can add $7k$ to x and subtract $19k$ from y for any integer k to get another solution. So, the set of all solutions is $\{(3 + 7k, -8 - 19k) : k \in \mathbb{Z}\}$.

Example: Find integers x, y such that $19x + 7y = 6$.

We do know that

$$\begin{aligned} 1 &= 3 \cdot 19 - 8 \cdot 7 \\ 6 &= 18 \cdot 19 - 48 \cdot 7 \end{aligned}$$

Then, we can similarly add $7k$ to x and subtract $19k$ from y for any integer k to get another solution. So, the set of all solutions is $\{(18 + 7k, -48 - 19k) : k \in \mathbb{Z}\}$.

Example: Find integers x, y such that $6x + 15y = 10$.

There are no integer solutions since $\gcd(6, 15) = 3$ does not divide 10. Generally if $\gcd(a, b)$ does not divide c , then there are no integer solutions to $ax + by = c$.

Proof: Let $d = \gcd(a, b)$. Then $d|a, d|b$ and $d|ax + by$ by bilinearity of divisibility. However, $d \nmid c$ by assumption, so there are no integer solutions to $ax + by = c$. $\square$

Theorem (3.1): Assume $a, b, c \in \mathbb{Z}$, and at least one of $a, b \neq 0$, and let $d = \gcd(a, b)$. Then,

- (1) The equation $ax + by = c$ has integer solutions iff $d|c$.
- (2) Additionally, if it has an integer solution, it has infinitely many integer solutions. In particular, if (x_0, y_0) is one integer solution, then the set of all integer solutions is given by

$$\{(x_0 + \frac{b}{d}k, y_0 - \frac{a}{d}k) : k \in \mathbb{Z}\}$$

Proof:

- (1) $\Rightarrow$: Let $ax + by = c$ have integer solutions. Then, since $d|a, d|b$, we have that $d|ax + by$ by bilinearity of divisibility. Thus, $d|c$.

$\Leftarrow$ Suppose $d|c$. Then, $c = d \cdot e$ for some integer e . Since d is a linear combination of a and b , we have that $d = ax_0 + by_0$ for some integers x_0, y_0 . Thus,

$$\begin{aligned} c &= d \cdot e \\ &= (ax_0 + by_0)e \\ &= a(ex_0) + b(ey_0) \end{aligned}$$

So, (ex_0, ey_0) is an integer solution to $ax + by = c$.

- (2) (This is my own proof) First, $\frac{b}{d}, \frac{a}{d} \in \mathbb{Z}$ by proposition 2.10. Now, let (x_0, y_0) be one integer solution to $ax + by = c$. Then, for any integer k , we have

$$\begin{aligned} a(x_0 + \frac{b}{d}k) + b(y_0 - \frac{a}{d}k) &= ax_0 + by_0 + ab\frac{k}{d} - ab\frac{k}{d} \\ &= c \end{aligned}$$

Thus, $(x_0 + \frac{b}{d}k, y_0 - \frac{a}{d}k)$ is an integer solution for any integer k . $\square$

Definition (Homogeneous LDE): A linear diophantine equation of the form $ax + by = 0$ is called a homogeneous linear diophantine equation.

Lemma: Suppose a, b are not both 0. Then

- (1) If $\gcd(a, b) = 1$, every solution to $ax + by = 0$ is of the form $(bk, -ak)$ for some integer k .
- (2) In general, if $d = \gcd(a, b)$, every solution to $ax + by = 0$ is of the form $(\frac{b}{d}k, -\frac{a}{d}k)$ for some integer k .

Proof:

- (1) Let a, b not both 0. Suppose $\gcd(a, b) = 1$. Let (x, y) be a solution to $ax + by = 0$. Then, $ax = -by$ and so $a|by$. Since $\gcd(a, b) = 1$, it follows that $a|y$. Thus, $y = ak$ for some integer k . Substituting back gives $ax = -bak$, so $x = -bk$. Thus, every solution is of the form $(bk, -ak)$ for some integer k .
- (2) Let a, b not both 0. Let $d = \gcd(a, b)$. Let (x, y) be a solution to $ax + by = 0$. Then, $ax = -by$ and so $a|by$. Since $\gcd(a, b) = d$, it follows that $\frac{a}{d}|\frac{b}{d}y$. Since $\gcd(\frac{a}{d}, \frac{b}{d}) = 1$ by proposition 2.10, it follows that $\frac{a}{d}|y$. Thus, $y = \frac{a}{d}k$ for some integer k . Substituting back gives $ax = -b\frac{a}{d}k$, so $x = -\frac{b}{d}k$. Thus, every solution is of the form $(\frac{b}{d}k, -\frac{a}{d}k)$ for some integer k .

9 Lecture 8

Theorem (3.1): Let $d = \gcd(a, b)$. Then

1. $ax + by = c$ has solutions iff $d|c$.
2. If (x_0, y_0) is one solution to $ax + by = c$, then the solution set is given by

$$x = x_0 + \frac{b}{d}t, y = y_0 - \frac{a}{d}t$$

where t is an integer

We proved this last time, along with a related lemma. I will now write Prof's proof of part 2.

Proof: (of part 2) Suppose (x_0, y_0) is some solution for $ax + by = c$. Then, we should show that the above format gives solutions, and additionally that they are all of the solutions. So

- Verify it is a solution:

$$\begin{aligned} a(x_0 + \frac{b}{d}t) + b(y_0 + \frac{a}{d}t) &= ax_0 + \frac{ab}{d}t + by_0 - \frac{ab}{d}t \\ &= ax_0 + by_0 \end{aligned}$$

which is a solution by assumption.

- Verify it is all the solutions: Let (x, y) be any solution to $ax + by = c$. Then,

$$\begin{aligned} a(x - x_0) + b(y - y_0) &= ax + by - (ax_0 + by_0) \\ &= c - c \\ &= 0 \end{aligned}$$

Thus, $(x - x_0, y - y_0)$ is a solution to the homogeneous linear diophantine equation $ax + by = 0$. By the previous lemma, there exists an integer t such that

$$(x - x_0, y - y_0) = (\frac{b}{d}t, -\frac{a}{d}t)$$

Thus,

$$(x, y) = (x_0 + \frac{b}{d}t, y_0 - \frac{a}{d}t)$$

which is of the desired form.

□

Note: Thus, to solve a linear diophantine equation

$$ax + by = c$$

, first ensure that $d = \gcd(a, b)$ divides c . If not, there are no solutions. If it does, then find one solution (x_0, y_0) using the extended euclidean algorithm. Then, the set of all solutions is given by

$$\{(x_0 + \frac{b}{d}t, y_0 - \frac{a}{d}t) : t \in \mathbb{Z}\}$$

Example: Solve $12x + 45y = 30$. Then, $\gcd(12, 45) = 3, 3|30$ so there are solutions

$$45 = 12(3) + 9$$

$$12 = 9(1) + 3$$

$$9 = 3(3) + 0$$

going back up we have

$$\begin{aligned} 3 &= 12 - 9(1) \\ &= 12 - (45 - 12(3)) \\ &= 4 \cdot 12 - 45 \end{aligned}$$

we can multiply by 10 to get $30 = 40 \cdot 12 - 10 \cdot 45$. Thus, one solution is $(40, -10)$. The set of all solutions is given by

$$\{(40 + 15t, -10 - 4t) : t \in \mathbb{Z}\}$$

9.1 3.2 - Postage Stamp Problem

Problem: You have an unlimited supply of 3 cent and 5 cent stamps. What amounts of postage can you make? Generalize this to a cent and b cent stamps, frequently where a, b are coprime.

The answer is obviously intuitively

$$ax + by = c$$

for some nonnegative integers x, y . We want to find the largest c such that there are no nonnegative integer solutions to this equation. Also, what are all of the values of c such that there are no nonnegative integer solutions to this equation?

Professor gives an example for postage stamps with values of 3 and 5 here, which is a trivially solved problem. The idea is that if you can find 3 consecutive values of c that can be achieved, then all values of c greater than the largest of those 3 must also be achievable. This is because you can add 3 to any of those 3 values to get the next value, and so on. Thus, you just need to find the largest c such that there are no 3 consecutive values of c that can be achieved.

In general, we can intuit from this that for the smallest of a, b , if we can find a consecutive values of c that can be achieved, then all values of c greater than the largest of those a must also be achievable. Thus, you just need to find the largest c such that there are no a consecutive values of c that can be achieved.

Example: Suppose in some fucked up version of football the only ways to score are with 3 or 7 points. What scores can be achieved?

Remember you just need 3 in a row that can be achieved and everything after is doable.

Score	Achievable?
0	Yes
1	No
2	No
3	Yes
4	No
5	No
6	Yes
7	Yes
8	No
9	Yes
10	Yes
11	No
12	Yes
13	Yes
14	Yes

and so on. We can see that the largest score that cannot be achieved is 11, and all scores greater than 11 can be achieved.

Proposition (3.3): Suppose $a > 1, b > 1$ are coprime. Then there are no nonnegative integers x, y such that $ax + by = ab - a - b$.

Note: The general idea behind this proof is that when you plug in any arbitrary integer solution to $ax + by = ab - a - b$, you get that $ax_0 + by_0 = 1$. Since $a, b > 1$, this forces x_0, y_0 to be nonnegative and sum to at most 1, none of which can satisfy the equation.

Proof: Since a, b are coprime, there exist integers by a previous theorem that $ax_0 + by_0 = 1$.

We know that $(ab - a - b)x_0, (ab - a - b)y_0$ is a solution to $ax + by = ab - a - b$. Since $a, b > 1, ab - a - b > 0$, and so for both to be positive we must have $x_0, y_0 \geq 0$.

Thus, $ax_0 + by_0 \geq 0$. However, since x_0, y_0 are integers, they must sum to at most 1, otherwise $ax_0 + by_0 \geq 2a + 2b > 1$. Thus, $0 \leq x_0 + y_0 \leq 1$. By integer properties, this means that either $x_0 = 0, y_0 = 0$ or one of them is 1 and the other is 0.

Then, if $x_0 = 0, y_0 = 0$, then $ax_0 + by_0 = 0 \neq 1$. If $x_0 = 1, y_0 = 0$, then $ax_0 + by_0 = a > 1$. If $x_0 = 0, y_0 = 1$, then $ax_0 + by_0 = b > 1$. But we need it to be identically 1 by assumption, a contradiction. $\square$

The professor gives a proof that requires memorizing some answer for x_0, y_0 , which is why I derived my own.

Proof: (by the professor) Consider the Diophantine Equation

$$ax + by = ab - a - b$$

Notice that $x_0 = -1, y_0 = a - 1$ is a solution, as

$$\begin{aligned} a(-1) + b(a - 1) &= -a + ab - b \\ &= ab - a - b \end{aligned}$$

Thus, the general solution of the Diophantine Equation is $x = -1 + bt, y = a - 1 - at$ for some integer t .

WLOG, suppose (x, y) is a solution such that $x \geq 0$. Since $x \geq 0$, we have that $-1 + bt \geq 0$. This

implies that $t > 0$, an integer, so $t \geq 1$. Then, since $y = a - 1 - at$, we have that $y \leq a - 1 - a = -1 < 0$. Thus, if $x \geq 0$, then $y < 0$. Similarly, if $y \geq 0$, then $x < 0$. Thus, there are no nonnegative integer solutions to the equation. $\square$

10 Lecture 9

Recall: The postage stamp problem asks, given a, b coprime positive integers, what is the largest integer c such that there are no nonnegative integer solutions to $ax + by = c$?

Proposition (3.4): Let a, b be coprime positive integers. If $n \geq ab - a - b + 1$ then there are nonnegative integers x, y , such that $ax + by = n$.

Proof: Let (x_0, y_0) be an integer solution to $ax + by = n$, which exists because $\gcd(a, b) = 1$ divides n . Then, the set of all integer solutions is given by

$$\{(x_0 + bk, y_0 - ak) : k \in \mathbb{Z}\}$$

Divide y_0 by a to get $y_0 = qa + r$ for some integer q and $0 \leq r < a$ by the division algorithm. Consider the solution (x_1, y_1) corresponding to $k = q$, which is given by

$$\begin{aligned}(x_1, y_1) &= (x_0 + bq, y_0 - ak) \\ &= (x_0 + bq, r)\end{aligned}$$

Thus, $0 \leq y_1 < a$. We need to show that $x_1 \geq 0$. Note that we have that $n \geq ab - a - b + 1$, and also that $y_1 \leq a - 1$ because $y_1 = r < a$. This implies $-y_1 \geq -(a - 1)$. Since $ax_1 + by_1 = n$, we have

$$\begin{aligned}x_1 &= \frac{n - by_1}{a} \\ &\geq \frac{ab - a - b + 1 - b(a - 1)}{a} \\ &= \frac{ab - a - b + 1 - ab + b}{a} \\ &= \frac{1 - a}{a} \\ &= -1 + \frac{1}{a}\end{aligned}$$

But because x_1 is an integer, it must be that $x_1 \geq 0$. Thus, there are nonnegative integer solutions to $ax + by = n$. $\square$

Corollary: From the above two propositions, the largest integer c such that there are no nonnegative integer solutions to $ax + by = c$ is $ab - a - b$. That is, for all $n \geq ab - a - b + 1$, there are nonnegative integer solutions to $ax + by = n$, and for $n = ab - a - b$, there are no nonnegative integer solutions to $ax + by = n$.

Proof: Compose the previous 2 proofs.

Example: The 2026 Postage Stamp Problem uses values of:

Stamp	Stamp Value
Regular	78
Postcard	61

First, we must have that they are coprime. Since $78 = 2 \cdot 3 \cdot 13$ and 61 is prime, they are coprime.

Then, under this problem, the largest amount of postage that cannot be made is $78 \cdot 61 - 78 - 61 = 4619$. For all amounts of postage greater than 4619, there are nonnegative integer solutions to $78x + 61y = n$.

Example: Find the smallest for stamp values of 6, 8. These two are not coprime, so we will divide the equation by their gcd (2). Then, we find the largest amount of postage that cannot be made with stamps of values 3, 4, which is $3 \cdot 4 - 3 - 4 = 5$. Then, we multiply this by the gcd (2) to get 10 as the largest amount of postage that cannot be made with stamps of values 6, 8. For all amounts of postage greater than 10, there are nonnegative integer solutions to $6x + 8y = n$.

10.1 QUIZ 1 CUTOFF HERE

10.2 Appendix A.3 - Well-Ordering Principle and Mathematical Induction

Theorem (Well-Ordering Principle): Every nonempty subset of $\mathbb{N}$ (the positive integers) has a least element.

Note: This only applies to the naturals and not the integers, since the integers can be negative and thus have no least element. Most notably the proof of the well-ordering principle uses the principle of succession, which is a property of the naturals specifically due to Peano succession axioms. You can also prove this utilizing the completeness of the reals, but once again this requires the existence of integers specifically in a half open interval.

Specifically it requires the completeness property, showing that all bottom-bounded subsets of the reals have a least upper bound, and because of the transitivity of $\subseteq$, for $A \subseteq \mathbb{N}$, A has an infimum called a , and then $(n - 1, n]$ must contain an element of A for all $n \geq a$, so there must be some least element of A .

Note: We will use this to give a non-constructive proof of the following theorem:

Theorem (2.12): Let a, b be integers, not both 0 and let $d = \gcd(a, b)$. Then, there exist integers x, y such that $ax + by = d$.

Note: Last time, we proved this using the Extended Euclidean Algorithm, which is a constructive proof. This time, we will give a non-constructive proof using the Well-Ordering Principle.

Proof: Let $S = \{ax + by : x, y \in \mathbb{Z}, ax + by > 0\}$ be the set of all positive linear combinations of a and b . Since a, b are not both 0, there exists some positive linear combination of a and b , so S is nonempty. By the Well-Ordering Principle, S has a least element, say d . We will show that $d = \gcd(a, b)$.

First, we show that $d|a$ and $d|b$. By the division algorithm, we can write $a = dq + r$ for some integers q, r with $0 \leq r < d$. Then,

$$\begin{aligned} r &= a - dq \\ &= a - q(ax_0 + by_0) \\ &= a(1 - qx_0) + b(-qy_0) \end{aligned}$$

where x_0, y_0 are integers such that $d = ax_0 + by_0$. Thus, r is a linear combination of a and b . If $r > 0$, then $r \in S$ and $r < d$, contradicting the minimality of d . Thus, $r = 0$ and so $d|a$. Similarly, we can show that $d|b$.

Professor starts a contradiction by stating suppose $r \neq 0$ and then doing the above, instead of taking cases.

Next, we show that if e is any common divisor of a and b , then $e|d$. Since $e|a$ and $e|b$, it follows that $e|ax + by$ for any integers x, y . In particular, since $d = ax_0 + by_0$, it follows that $e|d$. Since $d > 0$, $e \leq d$. Thus, d is the greatest common divisor of a and b . $\square$

Note: This argument shows that $\gcd(a, b)$ is the least positive linear combination of a and b .

Remark: Let $d = \gcd(a, b)$.

11 Lecture 10

Induction: Let $P(n)$ be a statement about a positive integer n . For instance, n is a product of primes. Then, if $P(1)$ is true, and for all $n \geq 1$, if $P(n)$ is true, then $P(n+1)$ is true, then $P(n)$ is true for all positive integers n .

The induction scheme expressed in the above is called weak induction. There is also strong induction, which states that if $P(1)$ is true, and for all $n \geq 1$, if $P(1), P(2), \dots, P(n)$ are all true, then $P(n+1)$ is true, then $P(n)$ is true for all positive integers n .

Note that this means that for strong induction, we only have to prove $n = 1$, but we can *assume* that $P(1), P(2), \dots, P(n)$ are all true to prove $P(n+1)$. For weak induction, we have to prove $n = 1$ and then for each $n \geq 1$, we have to prove that if $P(n)$ is true, then $P(n+1)$ is true. Thus, strong induction is often easier to use than weak induction.

I think we are allowed to do so because of the well-ordering principle, which states that every nonempty subset of the positive integers has a least element. If there is some n such that $P(n)$ is false, then there is a least such n . Since $P(1)$ is true, $n > 1$. Since n is the least such integer, $P(1), P(2), \dots, P(n-1)$ are all true. Thus, by the strong induction hypothesis, $P(n)$ is true, a contradiction. Thus, there are no such integers n such that $P(n)$ is false, and so $P(n)$ is true for all positive integers n .

Additionally, we can express them from a logic perspective:

- $P(1) \wedge \forall n \geq 1, P(n) \rightarrow P(n+1) \rightarrow \forall n \geq 1, P(n)$
- $P(1) \wedge \forall n \geq 1, (P(1) \wedge P(2) \wedge \dots \wedge P(n)) \rightarrow P(n+1) \rightarrow \forall n \geq 1, P(n)$

Example: Let's prove that $1 + 3 + 5 + \dots + (2n-1) = n^2$ for all $n \geq 1$. That is,

$$\sum_{k=1}^n (2k-1) = n^2$$

Proof: We will use weak induction. For the base case, $n = 1$, we have $1 = 1^2$, so the base case holds. Now, suppose that for some $n \geq 1$, we have $\sum_{k=1}^n (2k-1) = n^2$. Then,

$$\begin{aligned} \sum_{k=1}^{n+1} (2k-1) &= \sum_{k=1}^n (2k-1) + (2(n+1)-1) \\ &= n^2 + 2n + 1 \\ &= (n+1)^2 \end{aligned}$$

Thus, by the principle of weak induction, we have that $\sum_{k=1}^n (2k-1) = n^2$ for all $n \geq 1$.

Example: Prove that $n! \geq n \cdot 2^n$ for all $n \geq 6$.

Proof: By induction,

- Base: $n = 6$, $6! = 720 \geq 6 \cdot 2^6 = 384$, so the base case holds.
- Induction Step: Suppose that for some $n \geq 6$, we have $n! \geq n \cdot 2^n$. Then,

$$\begin{aligned} (n+1)! &= (n+1)n! \\ &\geq (n+1)n \cdot 2^n \\ &\geq (n+1) \cdot 2^{n+1} \end{aligned}$$

Thus, by the principle of weak induction, we have that $n! \geq n \cdot 2^n$ for all $n \geq 6$. □

Example: Prove that $3|(n^3 - n)$ for all $n \geq 0$

Proof: By induction,

- Base: $n = 0$, $3|0^3 - 0 = 0$, so the base case holds.
- Induction Step: Suppose that for some $n \geq 0$, we have $3|(n^3 - n)$. Then,

$$\begin{aligned}(n+1)^3 - (n+1) &= n^3 + 3n^2 + 3n + 1 - n - 1 \\ &= n^3 - n + 3n^2 + 3n \\ &= (n^3 - n) + 3(n^2 + n)\end{aligned}$$

Since $3|(n^3 - n)$ by the induction hypothesis, and $3|(3(n^2 + n))$ by bilinearity of divisibility, it follows that $3|((n^3 - n) + 3(n^2 + n))$. Thus, by the principle of weak induction, we have that $3|(n^3 - n)$ for all $n \geq 0$. $\square$

Note: Replace 3 with any prime p to get that $p|(n^p - n)$ for all $n \geq 0$. This is a special case of Fermat's Little Theorem, which states that if p is prime and a is an integer not divisible by p , then $a^{p-1} \equiv 1 \pmod{p}$.

Lemma (4.3): Every integer $n \geq 2$ can be written as a product of primes

$$n = p_1 p_2 \cdots p_k$$

Proof: We will use strong induction on n .

- Base: For $n = 2$, $2 = 2$ is a product of primes, so the base case holds.
- Inductive Step: Now assume every integer k with $2 \leq k \leq n$ can be written as a product of primes. We want to show that $n + 1$ can be written as a product of primes. If $n + 1$ is prime, then we are done. If $n + 1$ is composite, then there exist integers a, b such that $2 \leq a, b \leq n$ and $n + 1 = ab$. By the induction hypothesis, we can write $a = p_1 p_2 \cdots p_k$ and $b = q_1 q_2 \cdots q_m$ for some primes p_i, q_j . Thus,

$$\begin{aligned}n + 1 &= ab \\ &= (p_1 p_2 \cdots p_k)(q_1 q_2 \cdots q_m) \\ &= p_1 p_2 \cdots p_k q_1 q_2 \cdots q_m\end{aligned}$$

which is a product of primes. Thus, by the principle of strong induction, every integer $n \geq 2$ can be written as a product of primes. $\square$

12 Lecture 11

Recall last time we did 3 main things; the well ordering principle, induction, and strong induction.

All 3 of the above are equivalent to each other. This potentially means that you could rewrite one proof using the others!

12.1 4.1 - The Starting Point (of Unique Factorizations)

Theorem (4.1): Let p be a prime and a, b be integers such that $p|ab$. Then, $p|a$ or $p|b$.

Proof: We proved this above in the euclidean algorithm process

Corollary (4.2): Let p be a prime and $a_1, a_2, \dots, a_k$ be integers such that $p|a_1 a_2 \cdots a_k$. Then, there exists some i such that $p|a_i$.

Proof: We use induction on k .

Base: If $k = 2$ then this is just the previous theorem, so the base case holds. If $k = 1$, then $p|a_1$ so the base case holds.

IS: Now, suppose that for some $k \geq 2$, if $p|a_1a_2 \cdots a_k$ then there exists some i such that $p|a_i$. We want to show that if $p|a_1a_2 \cdots a_{k+1}$ then there exists some i such that $p|a_i$. If $p|a_{k+1}$, then we are done. If not, then since $p|a_1a_2 \cdots a_{k+1}$ and $p \nmid a_{k+1}$, it follows that $p|(a_1a_2 \cdots a_k)$ by bilinearity of divisibility. By the induction hypothesis, there exists some i such that $p|a_i$. $\square$

12.2 4.2 - The Fundamental Theorem of Arithmetic

Theorem: Every integer $n \geq 2$ can be written as a product of primes in a unique way, up to the order of the factors. That is, if

$$\begin{aligned} n &= p_1p_2 \cdots p_k \\ &= q_1q_2 \cdots q_m \end{aligned}$$

where p_i, q_j are primes, then $k = m$ and there exists a permutation σ of $\{1, 2, \dots, k\}$ such that $p_i = q_{\sigma(i)}$ for all i .

Proof: We already proved the existence of prime factorizations.

Uniqueness: First, observe that if p is prime, then $p = p$ is a prime factorization. This is the only way to write p as a product of primes, since if

$$p = p_1p_2 \cdots p_k$$

is a prime factorization, then $p|p_1p_2 \cdots p_k$, so there exists some i such that $p|p_i$. Since p_i is prime, it follows that $p = p_i$. Thus, the only prime factorization of a prime is itself.

Now, let's use strong induction on n .

- Base Case: at $n = 2$, 2 is prime, so the only prime factorization of 2 is 2 itself, so the base case holds.
- Inductive Step: Suppose k is a number such that all n such that $2 \leq n \leq k$ have a unique prime factorization. Suppose $k + 1$ has two prime factorizations

$$p_1p_2 \cdots p_m = q_1q_2 \cdots q_n$$

Then, $p_1|(q_1q_2 \cdots q_n)$, so there exists some i such that $p_1|q_i$. Since q_i is prime, it follows that $p_1 = q_i$. We can cancel p_1 and q_i from both sides to get

$$p_2 \cdots p_m = q_1 \cdots q_{i-1}q_{i+1} \cdots q_n$$

By the induction hypothesis, the remaining factors must also be equal up to a permutation. Thus, the prime factorization of $k + 1$ is unique.

12.3 EXAM 1 CUTOFF HERE

13 Lecture 12

Exam on Friday, the 27th. Covers up to Fundamental Theorem of Arithmetic.

HW due Wednesday

Note: Often, we will write the prime factorization of an integer n as

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

where all p_i are unique, and a_i are integers strictly greater than zero (eg ≥ 1)

Sometimes, we'll write a prime factorization of an integer $n \geq 2$ as follows:

$$2^{n_2} 3^{n_3} 5^{n_5} \dots$$

which resembles an "infinite product" where each prime shows up. Then, each n_i can be zero. Additionally we must then have that only finitely many n_p are nonzero.

Example: We know that $20 = 2^2 \cdot 5$. This is equal to

$$2^2 \cdot 3^0 \cdot 5^1 \cdot 7^0 \dots$$

In this case $n_2 = 2, n_3 = 0, n_5 = 1, n_p = 0$ for primes $p \geq 7$.

Remark: We can write $n = 1$ this way too, with $n_p = 0$ for all p .

Proposition (4.4): Let a, b be positive integers with prime factorizations

$$a = 2^{a_2} 3^{a_3} 5^{a_5} \dots \quad b = 2^{b_2} 3^{b_3} 5^{b_5} \dots$$

Then, $a|b$ iff $a_p \leq b_p$ for every prime p .

Proof: $\Rightarrow$ If $a|b$ then $b = ac$ for some integer $c \geq 1$. Write $c = 2^{c_2} 3^{c_3} 5^{c_5} \dots$. Then,

$$\begin{aligned} b &= ac \\ &= (2^{a_2} 3^{a_3} 5^{a_5} \dots) (2^{c_2} 3^{c_3} 5^{c_5} \dots) \\ &= 2^{a_2+c_2} 3^{a_3+c_3} 5^{a_5+c_5} \dots \end{aligned}$$

By the fundamental theorem of arithmetic, the prime factorization of b is unique, so $b_p = a_p + c_p$ for every prime p . Since $c_p \geq 0$, it follows that $a_p \leq b_p$ for every prime p .

$\Leftarrow$ If $a_p \leq b_p$ for every prime p , then $b_p - a_p \geq 0$, and we can write $b = 2^{b_2} 3^{b_3} 5^{b_5} \dots$ as

$$\begin{aligned} b &= 2^{a_2} 3^{a_3} 5^{a_5} \dots (2^{b_2-a_2} 3^{b_3-a_3} 5^{b_5-a_5} \dots) \\ &= a(2^{b_2-a_2} 3^{b_3-a_3} 5^{b_5-a_5} \dots) \end{aligned}$$

Since only finitely many a_p, b_p are nonzero, it follows that only finitely many $b_p - a_p$ are nonzero, so $2^{b_2-a_2} 3^{b_3-a_3} 5^{b_5-a_5} \dots$ is a well defined integer. Thus, $a|b$. $\square$

Proposition (4.5): Let $a = 2^{a_2} 3^{a_3} 5^{a_5} \dots$ and $b = 2^{b_2} 3^{b_3} 5^{b_5} \dots$ be positive integers. Then, let $d_p = \min(a_p, b_p)$ for every prime p . Then, $\gcd(a, b) = 2^{d_2} 3^{d_3} 5^{d_5} \dots$

Example: If $a = 720 = 2^4 \cdot 3^2 \cdot 5$ and $b = 6400 = 2^2 \cdot 5^2 \cdot 3^2 \cdot 7 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7$

Then, $d_2 = 2, d_3 = 2, d_5 = 1, d_7 = 0$, and all other $d_p = 0$, so $\gcd(a, b) = 2^2 \cdot 3^2 \cdot 5 = 180$.

Proof: Let $d_p = \min(a_p, b_p)$ for every prime p , and let $d = 2^{d_2} 3^{d_3} 5^{d_5} \dots$. We will show that $d = \gcd(a, b)$. By definition, $d_p \leq a_p$, and $d_p \leq b_p$ for every prime p , so by the previous proposition, $d|a$ and $d|b$. Thus, it is a common divisor.

Let e be an arbitrary common divisor. Then, $e = 2^{e_2} 3^{e_3} 5^{e_5} \dots$ for some nonnegative integers e_p . Since $e|a$ and $e|b$, it follows by the previous proposition that $e_p \leq a_p$ and $e_p \leq b_p$ for every prime p .

This implies that $e_p \leq \min(a_p, b_p) = d_p$ for every prime p , so by the previous proposition, $e|d$. Additionally, by def of divisibility $e \leq d$. Thus, d is the greatest common divisor of a and b . $\square$

Definition (LCM): The least common multiple of two positive integers a, b is the smallest positive integer m such that $a|m$ and $b|m$. We denote this as $\text{lcm}(a, b)$.

Proposition (4.5): (continued) If $a = 2^{a_2}3^{a_3}5^{a_5} \dots$ and $b = 2^{b_2}3^{b_3}5^{b_5} \dots$ are positive integers, then let $m_p = \max(a_p, b_p)$ for every prime p . Then, $\text{lcm}(a, b) = 2^{m_2}3^{m_3}5^{m_5} \dots$.

Proof: Let $m_p = \max(a_p, b_p)$ for every prime p , and let $m = 2^{m_2}3^{m_3}5^{m_5} \dots$. We will show that $m = \text{lcm}(a, b)$. By definition, $m_p \geq a_p$ and $m_p \geq b_p$ for every prime p , so by the previous proposition, $a|m$ and $b|m$. Thus, it is a common multiple.

Corollary: $\text{gcd}(a, b) \cdot \text{lcm}(a, b) = a \cdot b$ for all positive integers a, b .

Proof: Let $d_p = \min(a_p, b_p)$ and $m_p = \max(a_p, b_p)$ for every prime p . Then,

$$\begin{aligned} \text{gcd}(a, b) \cdot \text{lcm}(a, b) &= (2^{d_2}3^{d_3}5^{d_5} \dots)(2^{m_2}3^{m_3}5^{m_5} \dots) \\ &= 2^{d_2+m_2}3^{d_3+m_3}5^{d_5+m_5} \dots \end{aligned}$$

Since $d_p = \min(a_p, b_p)$ and $m_p = \max(a_p, b_p)$, it follows that $d_p + m_p = a_p + b_p$ for every prime p . Thus,

$$\begin{aligned} \text{gcd}(a, b) \cdot \text{lcm}(a, b) &= 2^{a_2+b_2}3^{a_3+b_3}5^{a_5+b_5} \dots \\ &= (2^{a_2}3^{a_3}5^{a_5} \dots)(2^{b_2}3^{b_3}5^{b_5} \dots) \\ &= a \cdot b \end{aligned}$$

□

Note: This also implies that

$$\text{lcm}(a, b) = \frac{a \cdot b}{\text{gcd}(a, b)}$$

Example: (of Prop): If $a = 720 = 2^4 \cdot 3^2 \cdot 5$ and $b = 6400 = 2^2 \cdot 5^2 \cdot 3^2 \cdot 7 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7$, then $\text{lcm}(a, b) = 2^4 3^2 5^2 \cdot 7 = 25200$.

14 Lecture 13

14.1 We forgot a thing from Chapter 2:

Apparently we forgot to prove a thing in chapter 2:

Proposition (2.17): Let a, b, c be integers with $\text{gcd}(a, b) = 1$. If $a|c, b|c$, then $ab|c$.

Proof: Since $a|c, b|c$, we have that $c = ae, c = bf$ for integers e, f . Since $\text{gcd}(a, b) = 1$, there are integers x, y such that

$$1 = ax + by$$

Multiplying by c gives

$$\begin{aligned} c &= acx + bcy \\ &= a(bf)x + b(ae)y \\ &= ab(fx + ey) \end{aligned}$$

Thus, $ab|c$.

□

14.2 5.1 - A Puzzle

Proposition (5.1): Let $k \geq 2$ be an integer, let m be a positive integer. Then, m is a k -th power of an integer iff all exponents in the prime factorization of m are divisible by k . That is, if

$$m = 2^{m_2} 3^{m_3} 5^{m_5} \dots$$

then m is a k -th power of an integer iff $k|m_p$ for every prime p .

Proof: $\Rightarrow$ If m is a k -th power of an integer, then there exists some integer n such that $m = n^k$. Write $n = 2^{n_2} 3^{n_3} 5^{n_5} \dots$. Then,

$$\begin{aligned} m &= n^k \\ &= (2^{n_2} 3^{n_3} 5^{n_5} \dots)^k \\ &= 2^{kn_2} 3^{kn_3} 5^{kn_5} \dots \end{aligned}$$

By the fundamental theorem of arithmetic, the prime factorization of m is unique, so $m_p = kn_p$ for every prime p . Since n_p is an integer, it follows that $k|m_p$ for every prime p .

$\Leftarrow$ If $k|m_p$ for every prime p , then there exist integers n_p such that $m_p = kn_p$ for every prime p . Then,

$$\begin{aligned} m &= 2^{m_2} 3^{m_3} 5^{m_5} \dots \\ &= 2^{kn_2} 3^{kn_3} 5^{kn_5} \dots \\ &= (2^{n_2} 3^{n_3} 5^{n_5} \dots)^k \end{aligned}$$

Thus, m is a k -th power of an integer. □

Puzzle: We got a limerick puzzle:

There was a young lady named Chris,
who, when asked her age, answered this
Two-thirds of its square,
Is a cube, I declare.
Now what was the age of the miss?

Let A be her age, eg

$$A = 2^{a_1} 3^{a_3} 5^{a_5} \dots$$

Two thirds of its square is

$$\begin{aligned} \frac{2}{3}A^2 &= \frac{2}{3}(2^{2a_2} 3^{2a_3} 5^{2a_5} \dots) \\ &= 2^{2a_2+1} 3^{2a_3-1} 5^{2a_5} \dots \end{aligned}$$

Then, this must be a perfect cube, eg $2a_2 + 1$ must be divisible by 3, $2a_3 - 1$ must be divisible by 3, and $2a_i$ where $i \geq 5$ must be divisible by 3.

One possibility is $a_2 = 1, a_3 = 2, a_i = 0$. This gives $A = 2^1 \cdot 3^2 = 18$. This is probably the most reasonable answer

Other possibilities? We have $3|2a_2 + 1, 3|2a_3 - 1, 3|2a_i$ for $i \geq 5$. Since $3|2a_i$, it follows that $3|a_i$ because the gcd of 3 and 2 is 1.

Thus, $a_i = 3k_i$ for some integer k_i .

Then, $3|(2a_2 + 1)$ implies that $2a_2 + 1 = 3k_2$ for some k_2 . Note this implies k_2 must be odd! We can then write $k_2 = 2m_2 + 1$ for some integer m_2 . Then, $a_2 = 3m_2 + 1$.

Similarly, $3|(2a_3 - 1)$ implies that $2a_3 - 1 = 3k_3$ for some integer k_3 . Note this implies k_3 must be odd! We can then write $k_3 = 2m_3 + 1$ for some integer m_3 . Then, $a_3 = 3m_3 + 2$.

Then,

$$A = 2^{3m_2+1} 3^{3m_3+2} \prod_{i \geq 5} p_i^{3k_i} = 18(2^{m_2} 3^{m_3} \prod_{i \geq 5} p_i^{k_i})^3$$

so the complete solution set is $18M^3$ where M is an integer. The next smallest age, in fact, is 144. She could also be 1152 years old!

14.3 5.2 - Irrationality Proofs

Definition (Rational): $r \in \mathbb{R}$ is rational if $\exists a, b \in \mathbb{Z}$ st $b \neq 0$ and $r = \frac{a}{b}$.

Definition (Irrational): $r \in \mathbb{R}$ is irrational if it is not rational.

The field of rational numbers is denoted $\mathbb{Q}$, and the field of irrational numbers is $\mathbb{R} \setminus \mathbb{Q}$.

Also, because it is a field, we have that $\mathbb{Q}$ is closed under addition, subtraction, multiplication, and division (except by zero).

Further, the field is dense, eg for any two rational numbers $r_1 < r_2$, there exists a rational number r such that $r_1 < r < r_2$. In fact, there are infinitely many rational numbers between any two rational numbers.

Proof: Let $r_1 = \frac{a}{b}, r_2 = \frac{c}{d}$ where $a, b, c, d \in \mathbb{Z}$ and $b, d \neq 0$. Then, since $r_1 < r_2$, it follows that $ad < bc$ (just cross multiply to see this).

Then, let $r = \frac{a+c}{b+d}$. We will show that $r_1 < r < r_2$. First,

$$\begin{aligned} r - r_1 &= \frac{a+c}{b+d} - \frac{a}{b} \\ &= \frac{(a+c)b - a(b+d)}{b(b+d)} \\ &= \frac{cb - ad}{b(b+d)} \end{aligned}$$

Since $ad < bc$, it follows that $cb - ad > 0$, so $r - r_1 > 0$, so $r > r_1$. Next,

$$\begin{aligned} r_2 - r &= \frac{c}{d} - \frac{a+c}{b+d} \\ &= \frac{c(b+d) - (a+c)d}{d(b+d)} \\ &= \frac{cb - ad}{d(b+d)} \end{aligned}$$

Since $ad < bc$, it follows that $cb - ad > 0$, so $r_2 - r > 0$, so $r < r_2$. Thus, there exists a rational number r such that $r_1 < r < r_2$. $\square$

Any terminating decimal is rational.

15 Lecture 14

Definition (Rational): If $r = \frac{a}{b}$, where a, b are integers with $b \neq 0$, then r is rational.

Notation:

- $\mathbb{Z}$ is the set of all integers
- $\mathbb{Q}$ is the set of all rational numbers
- $\mathbb{R}$ is the set of all real numbers
- $\mathbb{R} \setminus \mathbb{Q}$ is the set of all irrational numbers

Theorem: If $r \in \mathbb{Q}$, we can always write

$$r = \frac{a}{b}$$

where a, b are integers with $b > 0$ and $\gcd(a, b) = 1$.

Proof: Given any $r = \frac{a}{b}$ with $\gcd(a, b) = d$. We can write

$$\begin{aligned} r &= \frac{a}{b} \\ &= \frac{a/d}{b/d} \end{aligned}$$

and note that $a/d, b/d$ are integers with $b/d > 0$ and $\gcd(a/d, b/d) = 1$. □

Theorem (5.2): $\sqrt{2}$ is irrational.

Proof: On the contrary suppose it's rational. Then

$$\begin{aligned} \sqrt{2} &= \frac{a}{b} \\ 2 &= \frac{a^2}{b^2} \\ a^2 &= 2b^2 \end{aligned}$$

This shows that $2|a^2$. Since 2 is prime, it follows that $2|a$. Thus, $a = 2c$ for some $c \in \mathbb{Z}$. Then,

$$\begin{aligned} 2b^2 &= a^2 \\ &= (2c)^2 \\ &= 4c^2 \\ b^2 &= 2c^2 \end{aligned}$$

which implies that $2|b^2$, so $2|b$. Thus, a and b are both even, which contradicts the fact that $\gcd(a, b) = 1$. Thus, $\sqrt{2}$ is irrational. □

Theorem (5.3): Let $k \geq 2$ be an integer. Let n be a positive integer that is not a k -th power of an integer. Then, $\sqrt[k]{n}$ is irrational.

Proof: We will show the contrapositive,

The contrapositive states: If $\sqrt[k]{n}$ is rational, then n is a k -th power of an integer.

Let $\sqrt[k]{n} = \frac{a}{b}$ where a, b are integers with $b > 0$ and $\gcd(a, b) = 1$. Then,

$$\begin{aligned} n &= \left(\frac{a}{b}\right)^k \\ &= \frac{a^k}{b^k} \\ &= 2^{k(a_2-b_2)} 3^{k(a_3-b_3)} 5^{k(a_5-b_5)} \dots \end{aligned}$$

By the fundamental theorem of arithmetic, the prime factorization of n is unique, so $n = 2^{k(a_2-b_2)} 3^{k(a_3-b_3)} 5^{k(a_5-b_5)} \dots$. By the previous proposition, n is a k -th power of an integer. $\square$

Example: $\sqrt[3]{9}, \sqrt[5]{8}$ are irrational. $2 = \sqrt[4]{16}$ is rational.

15.1 5.3 - The Rational Root Theorem

Theorem (5.4 - Rational Root Theorem): Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ be a polynomial with integer coefficients, where $a_n \neq 0, a_0 \neq 0$. If $r = \frac{u}{v}$ is a reduced rational number such that $P(r) = 0$, then $u|a_0$ and $v|a_n$.

Proof: Since $P(r) = 0$, we have that

$$\begin{aligned} 0 &= a_n \left(\frac{u}{v}\right)^n + a_{n-1} \left(\frac{u}{v}\right)^{n-1} + \dots + a_1 \left(\frac{u}{v}\right) + a_0 \\ \implies \\ a_n u^n + a_{n-1} u^{n-1} v + \dots + a_1 u v^{n-1} + a_0 v^n &= 0 \end{aligned}$$

by multiplying both sides by v^n . Then,

$$\begin{aligned} a_0 v^n &= -(a_n u^n + a_{n-1} u^{n-1} v + \dots + a_1 u v^{n-1}) \\ &= -u(a_n u^{n-1} + a_{n-1} u^{n-2} v + \dots + a_1 v^{n-1}) \end{aligned}$$

Since u divides the right hand side, it must divide the left hand side. Since $\gcd(u, v) = 1$, it follows that $u|a_0$. Similarly,

$$\begin{aligned} a_n v^n &= -(a_{n-1} u^{n-1} v + \dots + a_1 u v^{n-1} + a_0 v^n) \\ &= -v(a_{n-1} u^{n-1} + \dots + a_1 u v^{n-2} + a_0 v^{n-1}) \end{aligned}$$

Since v divides the right hand side, it must divide the left hand side. Since $\gcd(u, v) = 1$, it follows that $v|a_n$. $\square$

Example: Use the fact that

$$p(x) = 2x^3 + x^2 - 9$$

has a rational root to find all of its complex roots. If $r = \frac{u}{v}$ is a rational root, then $u| -9$ and $v|2$. Thus, $u = \pm 1, \pm 3, \pm 9$ and $v = \pm 1, \pm 2$. This implies the possible roots as

$$\pm 1, \pm 3, \pm 9, \pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{9}{2}$$

Trying them all so we can see which one is a root, we see that $r = \frac{3}{2}$ is a root. Thus, $x - \frac{3}{2}$ is a factor, and

$$\begin{aligned} p(x) &= \left(x - \frac{3}{2}\right)(2x^2 + 4x + 6) \\ &= 2\left(x - \frac{3}{2}\right)(x^2 + 2x + 3) \end{aligned}$$

Then, we can use the quadratic formula to find the roots of the remaining quadratic. This is a way to factor a higher coefficient polynomial, basically!

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Recall last time we proved the rational root theorem, which states that if $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ is a polynomial with integer coefficients, where $a_n \neq 0, a_0 \neq 0$, and $r = \frac{u}{v}$ is a reduced rational number such that $P(r) = 0$, then $u|a_0$ and $v|a_n$.

Some applications

Example: $\sqrt{2}$ is irrational.

$\sqrt{2}$ is a root of the polynomial $p(x) = x^2 - 2$. If $\sqrt{2}$ were rational, then by the rational root theorem, the numerator of $\sqrt{2}$ in lowest terms would divide -2 , and the denominator would divide 1 . Thus, the only possible rational roots are $\pm 1, \pm 2$. None of these are roots of $p(x)$, so $\sqrt{2}$ is irrational.

Example: $\sqrt[k]{n}$ is irrational unless n is a k -th power of an integer.

$\sqrt[k]{n}$ is a root of the polynomial $p(x) = x^k - n$. If $\sqrt[k]{n}$ were rational, then by the rational root theorem, the numerator of $\sqrt[k]{n}$ in lowest terms would divide $-n$, and the denominator would divide 1 . Thus, the only possible rational roots are $\pm 1, \pm n$. None of these are roots of $p(x)$ unless n is a k -th power of an integer, so $\sqrt[k]{n}$ is irrational unless n is a k -th power of an integer.

Example: Take $p(x) = 78x^3 - 574x^2 + 310x - 45$. Say we know that the roots are approximately $0.26015\dots, 0.3451\dots, 6.40651\dots$. Are any of them rational?

Well, if any of them is rational, then the denominator is a divisor of 78 . So if we multiply by 78 , the root will become an integer. If we do that here, we get $20.2918, 27.0000, 499.7082$. Thus, 27 is an integer, so $0.3451\dots$ is a rational root. We can then factor the polynomial with it.

16.1 5.4 - Pythagorean Triples

Definition (Pythagorean Triple): A pythagorean triple is a triple of positive integers (a, b, c) such that $a^2 + b^2 = c^2$, eg integer solutions to the pythagorean theorem.

For instance, $(3, 4, 5)$ is a pythagorean triple since $3^2 + 4^2 = 5^2$. Also, $(5, 12, 13)$ is a pythagorean triple since $5^2 + 12^2 = 13^2$. Also, $(8, 15, 17)$ is a pythagorean triple since $8^2 + 15^2 = 17^2$.

Multiples too, eg $(6, 8, 10)$.

Notes:

- If (a, b, c) is a pythagorean triple, then so is (ka, kb, kc) for any positive integer k .
- If (a, b, c) is a pythagorean triple, and $d = \gcd(a, b, c)$, then $(\frac{a}{d}, \frac{b}{d}, \frac{c}{d})$ is a pythagorean triple.

Definition (Primitive Pythagorean Triple): A primitive pythagorean triple is a pythagorean triple (a, b, c) such that $\gcd(a, b, c) = 1$.

Question: Are there infinitely many primitive pythagorean triples? And can we generate them? Turns out the answer is yes for both

Fibonacci's Method: One way to do so:

- Start with any odd integer a .
- Square it, and then write it as a sum of two consecutive integer. That is, write $a^2 = k + (k + 1)$ for some integer k .

- Then, we have that $a^2 = k^2 + (k+1)^2 - (k^2 + (k+1)^2 - 2k(k+1)) = k^2 + (k+1)^2 - 2k(k+1) = k^2 + (k+1)^2 - 2k^2 - 2k = k^2 + (k+1)^2 - 2k^2 - 2k = (k+1)^2 - k^2$. Thus, $a^2 = (k+1)^2 - k^2$, so $(a, k, k+1)$ is a primitive pythagorean triple. (clear since the integers are consecutive so the gcd is 1)

Example:

- $a = 3$ yields $a^2 = 9 = 4 + 5$, so $k = 4$, so we get the pythagorean triple $(3, 4, 5)$.
- $a = 5$ yields $a^2 = 25 = 12 + 13$, so $k = 12$, so we get the pythagorean triple $(5, 12, 13)$.
- $a = 7$ yields $a^2 = 49 = 24 + 25$, so $k = 24$, so we get the pythagorean triple $(7, 24, 25)$.

Another Construction:

- Start with a multiple of 4, eg $a = 4k$
- Square it to $16k^2$ and then divide by 4 to get $4k^2$.
- Take the two odds on either side of $4k^2$, which are $4k^2 - 1$ and $4k^2 + 1$. Then, this is a primitive pythagorean triple since the three numbers are consecutive odd integers, so the gcd is 1.

Example:

- $a = 4$ yields $a^2 = 16$, dividing by 4 gives 4, so we take the two odds on either side of 4, which are 3 and 5, so we get the pythagorean triple $(4, 3, 5)$.
- $a = 8$ yields $a^2 = 64$, dividing by 4 gives 16, so we take the two odds on either side of 16, which are 15 and 17, so we get the pythagorean triple $(8, 15, 17)$.

Theorem (5.5): Let (a, b, c) be a primitive pythagorean triple. Then WLOG c is odd and exactly one of a, b is even. WLOG say b is even. Then, there exists rprime integers $m < n$, one even, one odd, such that

$$a = n^2 - m^2, b = 2mn, c = n^2 + m^2$$

Further any triple of this form is a primitive pythagorean triple.

Examples:

- $n = 3, m = 2$ gives $a = 9 - 4, b = 2 \cdot 3 \cdot 2, c = 9 + 4$, so we get the pythagorean triple $(5, 12, 13)$.

17 Lecture 16

We will prove 5.5. First, we will prove some related lemmas.

Lemma (5.6): Let $k \geq 2$ and let a, b be rprime in $\mathbb{Z}$ such that ab is a k -th power of an integer. Then, a and b are both k -th powers of integers.

Proof: Write the prime factorizations

$$a = p_1^{r_1} p_2^{r_2} \dots p_m^{r_m}, b = q_1^{s_1} q_2^{s_2} \dots q_n^{s_n}$$

Since a, b are rprime, $p_i \neq q_j$ for all i, j . Then,

$$ab = p_1^{r_1} p_2^{r_2} \dots p_m^{r_m} q_1^{s_1} q_2^{s_2} \dots q_n^{s_n}$$

is the prime factorization of ab into distinct primes. Since ab is a k -th power of an integer, all exponents are multiples of k . Thus, $k|r_i$ for all i , and $k|s_j$ for all j . Thus, a and b are both k -th powers of integers. $\square$

Lemma (5.7):

1. The square of an even number is divisible by 4
2. The square of an odd number is one more than a multiple of 8

Proof:

1. Let $a = 2k$ for some integer k . Then, $a^2 = 4k^2$, which is divisible by 4.
2. Let $a = 2k + 1$ for some integer k . Then, $a^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$. Since one of $k, k + 1$ is even, it follows that $4k(k + 1)$ is divisible by 8. Thus, a^2 is one more than a multiple of 8.

□

Recall:

Theorem (5.5): Let (a, b, c) be a primitive pythagorean triple. Then WLOG c is odd and exactly one of a, b is even. WLOG say b is even. Then, there exists rprime integers $m < n$, one even, one odd, such that

$$a = n^2 - m^2, b = 2mn, c = n^2 + m^2$$

Further any triple of this form is a primitive pythagorean triple.

Proof: Let's start by proving the last assertion. That is, suppose $a^2 = n^2 - m^2, b = 2mn, c = n^2 + m^2$ for some rprime integers $m < n$, one even, one odd. Then,

$$\begin{aligned} a^2 + b^2 &= (n^2 - m^2)^2 + (2mn)^2 \\ &= (n^4 - 2n^2m^2 + m^4) + (4m^2n^2) \\ &= n^4 - 2n^2m^2 + m^4 + 4m^2n^2 \\ &= n^4 + 2n^2m^2 + m^4 \\ &= (n^2 + m^2)^2 \end{aligned}$$

We will show that this is primitive. Suppose on the contrary that p is a prime such that $p|a, p|b, p|c$. Then, note that $c = n^2 + m^2$. Because one of m, n is even, this is odd. Thus, $p \neq 2$.

Then, p divides $a + c = (n^2 - m^2) + (n^2 + m^2) = 2n^2$ which implies $p|n^2 \implies p|n$.

Then, p divides $c - a = (n^2 + m^2) - (n^2 - m^2) = 2m^2$ which implies $p|m^2 \implies p|m$.

However, this is a contradiction because m, n are rprime. Thus, the triple is primitive.

Now, we will show the first assertion. Let (a, b, c) be an arbitrary primitive pythagorean triple. We will show that WLOG c is odd and exactly one of a, b is even.

If a, b are both even, then c^2 is divisible by 4, a contradiction. If a, b are both odd, then $a^2 \equiv 1 \pmod{8}$ and $b^2 \equiv 1 \pmod{8}$, so $c^2 = a^2 + b^2 \equiv 2 \pmod{8}$, which is a contradiction. Thus, exactly one of a, b is even. WLOG say b is even.

Then, $a^2 + c^2 = b^2$ is divisible by 4, so a, c are both even or both odd. If a, c are both even, then b^2 is divisible by 8, so b is divisible by 4, so a, b, c are all even, a contradiction. Thus, a, c are both odd, so c is odd.

Write $b = 2b_1$ for some integer b_1 . Then,

$$4b_1^2 = b^2 = a^2 - c^2 = (c - a)(c + a)$$

which gives

$$\begin{aligned} b_1^2 &= \frac{(c-a)(c+a)}{4} \\ &= \left(\frac{c-a}{2}\right) \left(\frac{c+a}{2}\right) \end{aligned}$$

These are both integers since c is odd and a is odd.

We claim that $\gcd(\frac{c-a}{2}, \frac{c+a}{2}) = 1$. Suppose on the contrary that p is a prime such that $p|\frac{c-a}{2}$ and $p|\frac{c+a}{2}$. Then, p divides their sum, which is c , and their difference, which is a . Thus, $p|a, p|c$. Since $b^2 = a^2 + c^2$, it follows that $p|b^2$, so $p|b$. This contradicts the fact that (a, b, c) is primitive. Thus, $\gcd(\frac{c-a}{2}, \frac{c+a}{2}) = 1$.

By the lemma 5.6, since $b_1^2 = (\frac{c-a}{2})(\frac{c+a}{2})$ and $\gcd(\frac{c-a}{2}, \frac{c+a}{2}) = 1$, it follows that $\frac{c-a}{2}$ and $\frac{c+a}{2}$ are both perfect squares. Thus, there exist integers m, n such that $\frac{c-a}{2} = m^2$ and $\frac{c+a}{2} = n^2$.

Then, $c = n^2 + m^2$ and $a = n^2 - m^2$. Since $\gcd(\frac{c-a}{2}, \frac{c+a}{2}) = 1$, it follows that $\gcd(m^2, n^2) = 1$, so $\gcd(m, n) = 1$.

Finally, since $b^2 = a^2 + c^2 = (n^2 - m^2)^2 + (n^2 + m^2)^2 = 4n^2m^2$, it follows that $b = 2mn$. Additionally since c is odd and $c = n^2 + m^2$, it follows that one of n, m is even and the other is odd. $\square$

18 Lecture 17

Last Time: We clasified all primitive solutions to $a^2 + b^2 = c^2$. That is, $a = n^2 - m^2, b = 2mn, c = n^2 + m^2$ for $n \geq m$, $\gcd(m, n) = 1$ for one even, one odd.

What about solutions to

$$a^k + b^k = c^k$$

for $k > 2$? Turns out, there are no solutions with positive integers a, b, c . This is Fermat's Last Theorem, eventually proven by Andrew Wiles in 1995.

Question: Find all primitive Pythagorean Triples of the form $(21, b, c)$, if any exist at all. Then, if such a triple exists, it must satisfy that

$$21 = n^2 - m^2, b = 2mn, c = n^2 + m^2$$

for some pair (n, m) as in the Theorem. We need to write 21 as a difference of two squares in order to achieve this result. This works with $n = 11, m = 10$, as $121 - 100 = 21$. Then, we have the triple $b = 2(10)(11)$, and $c = 11^2 + 10^2$, so $(21, 220, 221)$ is a triple.

Also, $n = 5, m = 2$ works, so we have 21, 20, 29.

18.1 5.5 - Difference of Squares

Given a positive integer m , how many ways (if any) can we write

$$m = x^2 - y^2$$

as a difference of squares, where x, y are integers?

Theorem (5.8): Let m be a positive integer. Then, m can be written as a difference of squares iff. m is odd or m is a multiple of 4.

Proof: $\Leftarrow$ First note that

$$(n+1)^2 - n^2 = 2n + 1$$

So, any odd number can be written as a difference of squares.

Next, note that

$$(n+1)^2 - (n-1)^2 = 4n$$

This shows that any multiple of 4 is a difference of two squares.

$\Rightarrow$ Suppose $m = x^2 - y^2$ is a difference of squares. Then,

$$m = (x-y)(x+y)$$

Note that the numbers $x-y, x+y$ are both either even or odd. Thus, suppose $m = uv$ and u, v have the same parity. Then, let $x = \frac{u+v}{2}$ and $y = \frac{u-v}{2}$. Then, $x^2 - y^2 = m$. Note that this happens only if u, v have the same parity, so m is either odd or a multiple of 4. $\square$

Note: So there is a 1-1 correspondence between factorizations $m = uv$, where u, v have the same parity and $u \geq v$, and the ways to write $m = x^2 - y^2$.

Example: Find all ways to write 21 as a difference of squares. Consider the factorizations of $21 = uv$.

- $u = 21, v = 1$ gives $21 = 11^2 - 10^2$
- $u = 7, v = 3$ gives $21 = 5^2 - 2^2$

Example: Find all ways to write 40 as a difference of squares. Consider the factorizations of $40 = uv$.

- $u = 40, v = 1$ gives no valid solution because they are not the same parity
- $u = 20, v = 2$ gives $40 = 11^2 - 9^2$
- $u = 10, v = 4$ gives $40 = 7^2 - 3^2$
- $u = 8, v = 5$ gives no valid solution because they are not the same parity.

Example: Find all ways to write 40 as a difference of squares. Consider the factorizations of $40 = uv$.

- $u = 40, v = 1$ gives no valid solution because they are not the same parity
- $u = 20, v = 2$ gives $40 = 11^2 - 9^2$
- $u = 10, v = 4$ gives $40 = 7^2 - 3^2$
- $u = 8, v = 5$ gives no valid solution because they are not the same parity.

Example: Find all primitive pythagorean triples with $a = 45$. Note that we want to find $45 = n^2 - m^2$ as part of the pythagorean triple process. So write 45 as a difference of 2 squares!

- $u = 45, v = 1$ gives $n = 23, m = 22$. This gives us the valid pythagorean triple $(45, 1012, 1013)$
- $u = 15, v = 3$ gives $n = 9, m = 6$. However, because these 2 are not relatively prime, we cannot get a primitive pythagorean triple from this. But they do give a normal one.

19 Lecture 18

Congruences

19.1 6.1 - Definitions and Examples

Let m be a positive integer.

Definition: For m a positive integer, two integers a, b are congruent modulo m , denoted $a \equiv b \pmod{m}$, if $m|a - b$.

Notation: $a \equiv b \pmod{m}$ means a and b are in the same residue class modulo m . That is,

$$\begin{aligned} a \equiv b \pmod{m} &\iff m|a - b \\ &\iff a - b = km \text{ for some integer } k \\ &\iff a = km + b \text{ for some integer } k \end{aligned}$$

Examples:

- $8 \equiv 0 \pmod{2}$
- $13 \equiv 1 \pmod{2}$
- $13 \equiv 1 \pmod{6}$
- $5 \equiv 17 \pmod{2}$

Proposition (6.2): $a \equiv b \pmod{m} \iff a = b + km$ for some integer k .

Proof: $\Rightarrow$ If $a \equiv b \pmod{m}$, then $m|a - b$, so $a - b = km$ for some integer k , so $a = b + km$.

$\Leftarrow$ If $a = b + km$ for some integer k , then $a - b = km$, so $m|a - b$, so $a \equiv b \pmod{m}$. □

Note: Note that

- $a \equiv 0 \pmod{2} \iff a$ is even
- $a \equiv 1 \pmod{2} \iff a$ is odd
- and every integer a is congruent to either 0 or 1 mod 2.

Similarly $a \equiv 0 \pmod{3}$ means a is a multiple of 3, $a \equiv 1 \pmod{3}$ means a is one more than a multiple of 3, and $a \equiv 2 \pmod{3}$ means a is two more than a multiple of 3.

Proposition (6.3): If a is an integer and m is a positive integer then there is a unique integer r with $0 \leq r \leq m - 1$ such that $a \equiv r \pmod{m}$. Then, r is called the least nonnegative residue of $a \pmod{m}$.

Proof: By the division algorithm, we can write $a = mq + r$ for some unique q, r with $0 \leq r \leq m - 1$. Then, $a \equiv r \pmod{m}$ by the previous proposition. Uniqueness follows from the fact that if $a \equiv r' \pmod{m}$ for some r' with $0 \leq r' \leq m - 1$, then $r \equiv r' \pmod{m}$, so $m|r - r'$. Since $0 \leq r, r' \leq m - 1$, it follows that $r = r'$. □

Proposition (6.4): Let a, b, c be integers and M is a positive integer. Then

1. $a \equiv a \pmod{m}$ (reflexivity)
2. if $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$ (symmetry)
3. if $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$ (transitivity)

This proposition also says that congruence modulo m is an equivalence relation on $\mathbb{Z}$.

Proof:

1. m divides $0 = a - a$, so $a \equiv a \pmod{m}$.
2. Since $a \equiv b \pmod{m}$, we know $m|a - b$, so $a - b = km$ for some integer k . Then, $b - a = -km$, so $m|b - a$, so $b \equiv a \pmod{m}$.
3. Since $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, we know $m|a - b$ and $m|b - c$, so $a - c = (a - b) + (b - c)$ is divisible by m , so $a \equiv c \pmod{m}$.

□

Note: It follows from everything we've done today that $a \equiv b \pmod{m} \iff a, b$ have the same remainder when divided by m .

We'll do lots of arithmetic with congruences (modular arithmetic), for instance $11 = 3 \equiv 2 \pmod{12}$, $30 + 75 \equiv 45 \pmod{60}$, sometimes called clock arithmetic

for the second one, observe that $30 + 75 \equiv 30 + 15 \pmod{60}$ since $75 \equiv 15 \pmod{60}$, and $30 + 15 = 45$.

Proposition (6.5): If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

1. $a + c \equiv b + d \pmod{m}$
2. $a - c \equiv b - d \pmod{m}$
3. $ac \equiv bd \pmod{m}$

Proof:

1. Since $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, we know $m|a - b$ and $m|c - d$, so $a + c - (b + d) = (a - b) + (c - d)$ is divisible by m , so $a + c \equiv b + d \pmod{m}$.
2. Since $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, we know $m|a - b$ and $m|c - d$, so $a - c - (b - d) = (a - b) - (c - d)$ is divisible by m , so $a - c \equiv b - d \pmod{m}$.
3. Since $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, we know $m|a - b$ and $m|c - d$, so $ac - bd = a(c - d) + (a - b)d$ is divisible by m , so $ac \equiv bd \pmod{m}$.

□

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Example: "Simplify" $100 + 4 \cdot 35 \pmod{33}$. Then,

$$100 + 4 \cdot 35 \equiv 1 + 4 \cdot 2 \pmod{33}$$

eg the least nonnegative residue of $100 + 4 \cdot 35 \pmod{33}$ is 9.

Example: $99^4 \pmod{101}$ is $-2 \pmod{101}$, so we have

$$99^4 \equiv (-2)^4 \pmod{101}$$

which is 16 $\pmod{101}$.

Corollary (6.6): If $a \equiv b \pmod{m}$, then $a^n \equiv b^n \pmod{m}$ for all positive integers n .

Proof: Use proposition 6.5's case 3 with induction.

Warning: If $a \equiv b \pmod{m}$ and $n \equiv k \pmod{m}$, then $a^n \equiv b^k \pmod{m}$ is not necessarily true. For instance, $2 \equiv 5 \pmod{3}$ and $1 \equiv 4 \pmod{3}$, but $2^1 = 2 \not\equiv 5^4 = 625 \pmod{3}$.

We'll address exponents when we talk about Fermat's/Euler's theorem.

Another consequence of Prop 6.5 is that we can manipulate congruences by adding/subtracting/multiplying both sides by the same integer. That is, if we want to find x such that

$$5x + 1 \equiv 8 \pmod{9}$$

we can just solve mod 9 to get $x \equiv 5$. That is,

$$\begin{aligned} 5x + 1 &\equiv 8 \implies 5x \equiv 7 \\ &\implies 10x \equiv 14 \\ &\implies x \equiv 5 \pmod{9} \end{aligned}$$

Warning: You cannot divide both sides of a congruence by the same integer in general. For instance, $2x \equiv 4 \pmod{6}$ has the set $x \equiv 2 \pmod{6}$ as ONE solution, but $x \equiv 5 \pmod{6}$ is another solution that we got rid of by dividing by 2.

Proposition (6.7): If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Proof: This follows from Proposition 6.8, which is

Proposition (6.8): If $ac \equiv bc \pmod{m}$, and $\gcd(c, m) = d$, then $a \equiv b \pmod{m/d}$ AND that $a \equiv b + (m/d)k \pmod{m}$ for some integer k such that $0 \leq k \leq d - 1$. Essentially, it's saying that we can divide both sides by c if we can also divide the modulus by the gcd of c and m .

Example: Suppose $10x \equiv 40 \pmod{55}$. then, $\gcd(10, 55) = 5$, so we can divide both sides by 5 to get $2x \equiv 8 \pmod{11}$. Since $\gcd(2, 11) = 1$, we can divide both sides by 2 to get $x \equiv 4 \pmod{11}$.

Proof: Suppose $ac \equiv bc \pmod{m}$ and $d = \gcd(c, m)$. Then, $ac = bc + km$ for some $k \in \mathbb{Z}$. Then, $ac - bc = km$, and $(a - b)c = km$. This implies $(a - b)\frac{c}{d} = k\frac{m}{d}$.

This shows that $m/d \mid \frac{c}{d}(a - b)$, and since $\gcd(\frac{c}{d}, \frac{m}{d}) = 1$, it follows that $m/d \mid a - b$, so $a \equiv b \pmod{m/d}$.

So, $a = b + u(m/d)$ for some integer u . Write $u = dq + r$ with $0 \leq r \leq d - 1$ as in the division algorithm. Then, $a = b + (dq + r)(m/d) = b + rm + qdm \equiv b + rm \pmod{m}$. Thus, $a \equiv b + rm \pmod{m}$ for some integer r such that $0 \leq r \leq d - 1$. $\square$

Note: Prime moduli are nice because they allow us to divide by any nonzero integer.

Proposition (6.10): If p is a prime and a, b are integers such that $ab \equiv 0 \pmod{p}$, then $a \equiv 0 \pmod{p}$ or $b \equiv 0 \pmod{p}$.

Proof: Suppose $ab \equiv 0 \pmod{p}$. If $a \equiv 0 \pmod{p}$, then we are done. If $a \not\equiv 0 \pmod{p}$, then $\gcd(a, p) = 1$, so we can divide both sides of the congruence by a to get $b \equiv 0 \pmod{p}$. $\square$

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Corollary (6.11): Let p be a prime. Then, the congruence $x^2 \equiv 1 \pmod{p}$ has only the solutions $x \equiv 1 \pmod{p}$ and $x \equiv -1 \pmod{p}$.

Proof: Suppose $x^2 \equiv 1 \pmod{p}$. Then, $x^2 - 1 \equiv 0 \pmod{p}$, so $(x - 1)(x + 1) \equiv 0 \pmod{p}$. By the previous proposition, $x \equiv 1 \pmod{p}$ or $x \equiv -1 \pmod{p}$. $\square$

Example: Suppose $x^2 \equiv 1 \pmod{8}$. Then, we have the four solutions $x \equiv 1 \pmod{8}$, $x \equiv 3 \pmod{8}$, $x \equiv 5 \pmod{8}$, and $x \equiv 7 \pmod{8}$.

Example: Suppose $x^2 \equiv 1 \pmod{15}$. Then, we have the four solutions $x \equiv 1 \pmod{15}$, $x \equiv 4 \pmod{15}$, $x \equiv 11 \pmod{15}$, and $x \equiv 14 \pmod{15}$.

This is because $x^2 \equiv 1 \pmod{15}$ is equivalent to $x^2 \equiv 1 \pmod{3}$ and $x^2 \equiv 1 \pmod{5}$. The first congruence has the solutions $x \equiv 1 \pmod{3}$ and $x \equiv -1 \pmod{3}$, and the second congruence has the solutions $x \equiv 1 \pmod{5}$ and $x \equiv -1 \pmod{5}$. By the Chinese Remainder Theorem, we can combine these to get the four solutions listed above.

21.1 6.2 - Modular Exponentiation

For a lot of cryptography, it's important to calculate $a^n \pmod{m}$ for large a, n, m .

Example: Let's evaluate $5^{199} \pmod{101}$. We can keep squaring and reducing $\pmod{101}$, eg

$$\begin{aligned} 5^2 &\equiv 25 \pmod{101} \\ 5^4 &\equiv 25^2 \equiv 625 \equiv 19 \pmod{101} \\ \dots 5^{128} &\equiv 24 \pmod{101} \end{aligned}$$

Then, $199 = 128 + 64 + 4 + 2 + 1$ and so

$$5^{199} = 5^{128} 5^{64} 5^4 5^2 5^1 \equiv 24 \cdot 78 \cdot 19 \cdot 25 \cdot 5 \equiv 81 \pmod{101}$$

This is pretty efficient as it turns out! We just need to compute the mods in the powers of 2 and then break them down.

21.2 6.3 - Divisibility Tests

Is 8394 divisible by 2? Is 9125? It turns out that " n is divisible by 2 iff its last digit is divisible by 2", as we all know. The decimal expansion of 8394 is as follows:

$$\begin{aligned} 8394 &= 8 \cdot 10^3 + 3 \cdot 10^2 + 9 \cdot 10^1 + 4 \cdot 10^0 && \equiv 0 + 0 + 0 + 4 \pmod{2} \\ &\equiv 4 \pmod{2} \\ &\equiv 0 \pmod{2} \end{aligned}$$

Proposition (6.13): n is congruent to its last 2 digits $\pmod{4}$, and is congruent to its last 3 digits $\pmod{8}$.

Proof: This follows from the fact that $10^2 \equiv 0 \pmod{4}$ and $10^3 \equiv 0 \pmod{8}$.

Proposition (6.14): n is congruent $\pmod{3}$ (or $\pmod{9}$) to the sum of its digits. Consequently, n is divisible by 3 (or 9) iff the sum of its digits is divisible by 3 (or 9).

Proof: If $k \geq 1$, then $10^k \equiv 1 \pmod{3}$ and $10^k \equiv 1 \pmod{9}$. Thus, if $n = a_k 10^k + a_{k-1} 10^{k-1} + \dots + a_1 10 + a_0$ is the decimal expansion of n , then

$$\begin{aligned} n &= a_k 10^k + a_{k-1} 10^{k-1} + \dots + a_1 10 + a_0 \\ &\equiv a_k + a_{k-1} + \dots + a_1 + a_0 \pmod{3} \end{aligned}$$

and

$$\begin{aligned} n &= a_k 10^k + a_{k-1} 10^{k-1} + \dots + a_1 10 + a_0 \\ &\equiv a_k + a_{k-1} + \dots + a_1 + a_0 \pmod{9} \end{aligned}$$

Thus, n is divisible by 3 (or 9) iff the sum of its digits is divisible by 3 (or 9). □

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Last Time: Divisibility tests: we did 3, 9, etc.

Lastly, consider divisibility by 11. We know that $10 \equiv -1 \pmod{11}$, so $10^2 \equiv 1 \pmod{11}$, and generally

$$10^k \equiv (-1)^k \pmod{11}$$

Suppose $n = a_k 10^k + a_{k-1} 10^{k-1} + \cdots + a_1 10 + a_0$ is the decimal expansion of n . Then,

$$\begin{aligned} n &= a_k 10^k + a_{k-1} 10^{k-1} + \cdots + a_1 10 + a_0 \\ &\equiv a_k (-1)^k + a_{k-1} (-1)^{k-1} + \cdots + a_1 (-1) + a_0 \pmod{11} \end{aligned}$$

Thus, n is divisible by 11 iff the alternating sum of its digits is divisible by 11. For instance, 123456 is divisible by 11 iff $1 - 2 + 3 - 4 + 5 - 6 = -3$ is divisible by 11, which is not the case.

Proposition (6.16): n is congruent mod 11 to the alternating sum of its digits. Consequently, n is divisible by 11 iff the alternating sum of its digits is divisible by 11.

Proof: This follows from the fact that $10^k \equiv (-1)^k \pmod{11}$. □

22.1 6.4 - Linear Convergence

Given integers a, b, m we want to solve the linear congruence

$$ax \equiv b \pmod{m}$$

for x . We want $(\text{mod } m)$ solutions, so we usually give least nonnegative residues as solutions.

Example: We could try all the possibilities, eg for

$$3x \equiv 2 \pmod{5}$$

We can try $x = 0, 1, 2, 3, 4$ to see if any of these work. We find that $x = 4$ works, so the solution is $x \equiv 4 \pmod{5}$.

Example: We can try all the possibilities for

$$4x \equiv 2 \pmod{10}$$

has solutions $x = 3, 8, \dots$

Example: We can try all the possibilities for

$$4x \equiv 3 \pmod{10}$$

has no solutions.

To study linear congruences, we will convert them to linear Diophantine equations. That is, we want to solve

$$\begin{aligned} ax \equiv b \pmod{m} &\iff ax = b + km \text{ for some integer } k \\ &\iff ax - km = b \end{aligned}$$

for some integer k . This is a linear Diophantine equation in the variables x and k . We know how to solve these from before, so we can solve linear congruences by converting them to linear Diophantine equations.

Recall that the diophantine equation

$$ax - my = b$$

has solutions iff $d = \gcd(a, m) | b$. If $d | b$, then there are infinitely many solutions, and the solutions are of the form

$$(x, y) = (x_0, y_0) + t \left(\frac{m}{d}, \frac{a}{d} \right)$$

for some integer t , where (x_0, y_0) is a particular solution.

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Theorem (6.18): Let $m > 0, a \not\equiv 0 \pmod{m}$. Then, the congruence $ax \equiv b \pmod{m}$ has a solution iff $d = \gcd(a, m)$ divides b . Additionally there are exactly d distinct solutions up to $\pmod{m}$.

Moreover if x_0 is one solution, then the collection of all solutions $\pmod{m}$ are given by

$$x \equiv x_0 + \frac{m}{d} \cdot k \pmod{m}, 0 \leq k < d$$

The solution x_0 can be taken to be any integer solution of

$$\frac{a}{d}x \equiv \frac{b}{d} \pmod{\frac{m}{d}}$$

Proof: We know that

$$ax \equiv b \pmod{m} \iff \exists y ax - my = b$$

Then, using Theorem 3.1, the diophantine equation has solutions iff $d | b$, and so the congruence has solutions iff that holds.

If $d | b$, then all solutions of the Diophantine equation have the form

$$\begin{aligned} x &= x_0 - \left(\frac{-m}{d} \right) k, y = y_0 + \frac{a}{d} k \\ x &= x_0 + \frac{m}{d} k, y = y_0 + \frac{a}{d} k \end{aligned}$$

Here, (x_0, y_0) is one solution of the Diophantine equation. Then, all solutions of $ax \equiv b \pmod{m}$ have the form

$$x \equiv x_0 + \frac{m}{d} k \pmod{m}$$

for some integer k (and x_0 is a solution of $ax \equiv b \pmod{m}$). Now, it only remains to show that $0 \leq k < d$. Note that

$$x_0 + \frac{m}{d} k \equiv x_0 + \frac{m}{d} (k + d) \pmod{m}$$

so we can reduce $k \pmod{d}$ to get $0 \leq k < d$.

Lastly, if x_0 is any solution of $\left(\frac{a}{d} x_0 \right) \equiv \frac{b}{d} \pmod{\frac{m}{d}}$, then

$$\begin{aligned} \frac{a}{d} x_0 \equiv \frac{b}{d} \pmod{\frac{m}{d}} &\iff ax_0 = b + mt \text{ for some integer } t \\ &\iff ax_0 \equiv b \pmod{m} \end{aligned}$$

Corollary (6.19): If $\gcd(a, m) = 1$, then the congruence $ax \equiv b \pmod{m}$ has a unique solution $\pmod{m}$.

Example: $4x \equiv 3 \pmod{10}$ has no solutions since $\gcd(4, 10) = 2$ does not divide 3.

Example: $4x \equiv 2 \pmod{10}$ has solutions since $\gcd(4, 10) = 2$ divides 2. We can divide both sides by 2 to get $2x \equiv 1 \pmod{5}$.

$x_0 = 3$ solves this. All solutions must thus have the form

$$\begin{aligned} x &\equiv 3 + (10/2)k \pmod{10} \\ x &\equiv 3 + 5k \pmod{10}, 0 \leq k < 2 \end{aligned}$$

This gives us the two solutions $x \equiv 3 \pmod{10}$ and $x \equiv 8 \pmod{10}$.

Example: $3x \equiv 12 \pmod{18}$ has solutions because $\gcd(3, 18) = 3$ divides 12.

Clearly 4 is a solution, and all solutions must have the form

$$\begin{aligned} x &\equiv 4 + (18/3)k \pmod{18} \\ x &\equiv 4 + 6k \pmod{18}, 0 \leq k < 3 \end{aligned}$$

This gives us the three solutions $x \equiv 4 \pmod{18}$, $x \equiv 10 \pmod{18}$, and $x \equiv 16 \pmod{18}$.

Example: $100x \equiv 11 \pmod{29}$. First, $100 \equiv 13 \pmod{29}$. Thus, we are solving $13x \equiv 11 \pmod{29}$.

Since $\gcd(13, 29) = 1$, there is a unique solution $\pmod{29}$. Then, we will try to solve

$$\begin{aligned} 13x \equiv 11 \pmod{29} &\iff 13x = 11 + 29k \text{ for some integer } k \\ &\iff 13x - 29k = 11 \end{aligned}$$

Use the extended Euclidean algorithm to solve this. We have

$$\begin{aligned} 29 &= 13 \cdot 2 + 3 \\ 13 &= 3 \cdot 4 + 1 \\ 3 &= 1 \cdot 3 + 0 \end{aligned}$$

Then, we can back substitute to get

$$\begin{aligned} 1 &= 13 - 3 \cdot 4 \\ &= 13 - 4(29 - 13 \cdot 2) \\ &= 9 \cdot 13 - 4 \cdot 29 \end{aligned}$$

Thus, $11 = 11 \cdot 1 = 11(9 \cdot 13 - 4 \cdot 29) = 99 \cdot 13 - 44 \cdot 29$. This gives us the solution $x = 99 \equiv 12 \pmod{29}$. That is, $100 \cdot 12 \equiv 11 \pmod{29}$.

23.1 Multiplicative Inverses

Definition (Multiplicative Inverse): We say that integers a, b are multiplicative inverses $\pmod{m}$ if $ab \equiv 1 \pmod{m}$.

Then we will write that

$$a^{-1} \equiv b \pmod{m}, \text{ or } b^{-1} \equiv a \pmod{m}$$

Example: $(13)(9) \equiv 1 \pmod{29}$ so $13^{-1} \equiv 9 \pmod{29}$ and $9^{-1} \equiv 13 \pmod{29}$.

Example: In $(\text{mod } 7)$, $2 \cdot 4 \equiv 1 \pmod{7}$ so $2^{-1} \equiv 4 \pmod{7}$ and $4^{-1} \equiv 2 \pmod{7}$.

Note that the inverse of $a \pmod{m}$ is a solution to

$$ax \equiv 1 \pmod{m}$$

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24.1 Multiplicative Inverses $\pmod{m}$

Recall that for integers a, b and a positive integer m , we say that a and b are multiplicative inverses $\pmod{m}$ if $ab \equiv 1 \pmod{m}$.

Example: $(\text{mod } 1)0$

$$1 \cdot 1 \equiv 1 \pmod{10} \implies 1^{-1} \equiv 1 \pmod{10}$$

$$3 \cdot 7 \equiv 1 \pmod{10} \implies 3^{-1} \equiv 7 \pmod{10}$$

$$9 \cdot 9 \equiv 1 \pmod{10} \implies 9^{-1} \equiv 9 \pmod{10}$$

here, 0,2,4,5,6,8 do not have inverses $\pmod{10}$. This is because $\gcd(0, 10) = 10$, $\gcd(2, 10) = 2$, $\gcd(4, 10) = 2$, $\gcd(5, 10) = 5$, $\gcd(6, 10) = 2$, and $\gcd(8, 10) = 2$ are not equal to 1.

Note: The integers that have multiplicative inverses $\pmod{m}$ are precisely those that are rprime to m .

Corollary (6.21): The integer a has an inverse $\pmod{m}$ iff $\gcd(a, m) = 1$.

Furthermore, in this case, the inverse is unique $\pmod{m}$.

Proof: In order to find the inverse of $a \pmod{m}$, we need to solve

$$ax \equiv 1 \pmod{m}$$

By Theorem 6.18, this has a solution iff $\gcd(a, m) = 1$.

Then, suppose x_0 is a solution. Then, all solutions have the form

$$\begin{aligned} x &\equiv x_0 + (m/1)k \pmod{m} \\ x &\equiv x_0 \pmod{m} \end{aligned}$$

so the solution is unique $\pmod{m}$. □

Example: In a previous example we found that $13(9) + 29(-4) = 1$, so $13(9) \equiv 1 \pmod{29}$, so $13^{-1} \equiv 9 \pmod{29}$.

Example: Find the inverse of $49 \pmod{100}$. We can use the extended Euclidean algorithm to find integers x, y such that $49x - 100y = 1$. We have

$$100 = 49 \cdot 2 + 2$$

$$49 = 2 \cdot 24 + 1$$

$$2 = 1 \cdot 2 + 0$$

Then, we can back substitute to get

$$\begin{aligned} 1 &= 49 - 2 \cdot 24 \\ &= 49 - 24(100 - 49 \cdot 2) \\ &= 49 \cdot 49 - 24 \cdot 100 \end{aligned}$$

Thus, $49(49) \equiv 1 \pmod{100}$, so $49^{-1} \equiv 49 \pmod{100}$.

Additionally, this inverse is unique $\pmod{100}$, so the only solution to $49x \equiv 1 \pmod{100}$ is $x \equiv 49 \pmod{100}$.

Suppose $x + 1|m$. Then, is the inverse of $x \pmod{m}$ always equal to itself if $\gcd(x, m) = 1$? For instance, $49 + 1 = 50|100$, and $49^{-1} \equiv 49 \pmod{100}$.

No, because $x = 2, m = 21$ is a counterexample. But what if $\frac{m}{x+1}|x - 1$? That is, if $x + 1$ divides m , and the ratio $\frac{m}{x+1}$ divides $x - 1$.

Proof: Suppose $x + 1|m$, and $\gcd(x, m) = 1$. (That is, x has an inverse $\pmod{m}$). Additionally, let $\frac{m}{x+1}|x - 1$. That is, $x - 1 = k \cdot \frac{m}{x+1}$ for some integer k .

Then, $m = k(x + 1)$ for some integer k by assumption. Then, we aim to show $x^2 \equiv 1 \pmod{m}$.

Note that $x^2 - 1 = (x - 1)(x + 1)$. Note that $x + 1|(x - 1)(x + 1)$, so $k(x + 1)|k(x - 1)(x + 1)$, so $m|k(x - 1)(x + 1)$, so $m|k(x^2 - 1)$. We aim to show that $k|(x - 1)$.

However, this is immediate since $x - 1 = k \cdot \frac{m}{x+1}$. Thus, $k|(x - 1)$, so $m|k(x^2 - 1)$ implies $m|(x^2 - 1)$, so $x^2 \equiv 1 \pmod{m}$. Thus, $x^{-1} \equiv x \pmod{m}$. $\square$

Furthermore, all families $x = kt + 1, m = k(kt + 2)$ for some $k > 0, t$ satisfies this. For example, if $k = 2, t = 3$, then $x = 2 * 3 + 1 = 7$ and $m = 2(6 + 2) = 16$. Then, $7^{-1} \equiv 7 \pmod{16}$.

Note: Using the inverse to solve $ax \equiv b \pmod{m}$: If a has an inverse $\pmod{m}$, then

$$\begin{aligned} ax &\equiv b \pmod{m} \\ a^{-1}ax &\equiv a^{-1}b \pmod{m} \\ x &\equiv a^{-1}b \pmod{m} \end{aligned}$$

is the unique solution $\pmod{m}$.

24.2 6.5 - Chinese Remainder Theorem

1700 years ago, Sun Tzu asks for a number x that leaves a remainder of 2 when divided by 3, a remainder of 3 when divided by 5, and a remainder of 2 when divided by 7. That is, we want to solve the system of congruences

$$\begin{aligned} x &\equiv 2 \pmod{3} \\ x &\equiv 3 \pmod{5} \\ x &\equiv 2 \pmod{7} \end{aligned}$$

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Goal: We aim to find all c such that $x \equiv c \pmod{mn}$, where

$$\begin{aligned} x &\equiv a \pmod{m} \\ x &\equiv b \pmod{n} \end{aligned}$$

Here's an approach that generalizes well:

Example: Take

$$\begin{cases} x \equiv 1 \pmod{3} \\ x \equiv 5 \pmod{7} \end{cases}$$

The first congruence says that $x = 1 + 3t$ for some t . Let's insert this into the second congruence, eg

$$\begin{aligned}x &\equiv 5 \pmod{7} \\1 + 3t &\equiv 5 \pmod{7} \\3t &\equiv 4 \pmod{7} \\t &\equiv 6 \pmod{7}\end{aligned}$$

so $t = 6 + 7u$ for some u . Then, $x = 1 + 3t = 1 + 3(6 + 7u) = 19 + 21u$. Thus, the solutions to the system of congruences are $x \equiv 19 \pmod{21}$.

Example: Solve

$$\begin{cases} x \equiv 1 \pmod{2} \\ 2x \equiv 1 \pmod{5} \\ x \equiv 7 \pmod{9} \end{cases}$$

Note that this system yields

$$\begin{cases} x \equiv 1 \pmod{2} \\ x \equiv 3 \pmod{5} \\ x \equiv 7 \pmod{9} \end{cases}$$

Then, $x \equiv 1 \pmod{2} \implies x = 1 + 2t$ for some t . Inserting this into the second congruence gives us

$$\begin{aligned}1 + 2t &\equiv 3 \pmod{5} \\2t &\equiv 2 \pmod{5} \\t &\equiv 1 \pmod{5}\end{aligned}$$

so $t = 1 + 5u$ for some u . Then, $x = 1 + 2t = 1 + 2(1 + 5u) = 3 + 10u$. Inserting this into the third congruence gives us

$$\begin{aligned}3 + 10u &\equiv 7 \pmod{9} \\10u &\equiv 4 \pmod{9} \\u &\equiv 4 \pmod{9}\end{aligned}$$

so $u = 4 + 9v$ for some v . Then, $x = 3 + 10u = 3 + 10(4 + 9v) = 43 + 90v$. Thus, the solutions to the system of congruences are $x \equiv 43 \pmod{90}$.

Theorem (6.22): (Chinese Remainder Theorem, 1st version): If m and n are relatively prime, then the system of congruences

$$\begin{aligned}x &\equiv a \pmod{m} \\x &\equiv b \pmod{n}\end{aligned}$$

has a unique solution $\pmod{mn}$.

Note: We know that

$$\begin{aligned}x &= a + mt \text{ for some integer } t \\ \implies a + mt &\equiv b \pmod{n} \\ \implies mt &\equiv b - a \pmod{n} \\ \implies t &\equiv m^{-1}(b - a) \pmod{n}\end{aligned}$$

Thus, t is uniquely determined $\pmod n$, so $x = a + mt$ is uniquely determined $\pmod{mn}$.

Proof: (Existence): Let t be an integer with the property that t satisfies $mt \equiv b - a \pmod n$. Since $\gcd(m, n) = 1$, m has an inverse $\pmod n$, so such a t exists.

Let $x = a + mt$. Then, we claim x is a solution of the system. Note that $\pmod m$, $x \equiv a + mt \equiv a \pmod m$. Also, $\pmod n$, $x \equiv a + mt \equiv a + (b - a) \equiv b \pmod n$. Thus, x is a solution of the system.

(Uniqueness): Suppose x_1, x_2 are two solutions. Then,

$$x_1 \equiv a \pmod m$$

$$x_2 \equiv a \pmod m$$

$$x_1 \equiv b \pmod n$$

$$x_2 \equiv b \pmod n$$

But this tells us that

$$x_1 \equiv x_2 \pmod m$$

$$\implies m \mid x_1 - x_2$$

and similarly $n \mid x_1 - x_2$. Since $\gcd(m, n) = 1$, it follows that $mn \mid x_1 - x_2$, so $x_1 \equiv x_2 \pmod{mn}$ by Prop 2.17. $\square$

Theorem: (Chinese Remainder Theorem, 2nd version): Let $m_1, m_2, \dots, m_k$ be pairwise relatively prime positive integers. Let $M = m_1 m_2 \cdots m_k$. Then, the system of congruences

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$\vdots$$

$$x \equiv a_k \pmod{m_k}$$

has a unique solution $\pmod M$.

Proof: We will prove this by induction on k , the number of congruences.

Note: Also, the CRT says that a congruence with a composite modulus is equivalent to a system of congruences with rprime moduli, which are usually the factors of the composite moduli.

Example: Suppose we have the problem $x \equiv 33 \pmod{90}$. Then, we can decompose this to

$$x \equiv 33 \pmod 2$$

$$x \equiv 33 \pmod 5$$

$$x \equiv 33 \pmod 9$$

, which is in turn equivalent to

$$x \equiv 1 \pmod 2$$

$$x \equiv 3 \pmod 5$$

$$x \equiv 6 \pmod 9$$

26 Lecture 25

Exam on Monday, up to and including 6.5.

Example: Find all $(\text{mod } 55)$ solutions to $x^2 \equiv 1 \pmod{55}$.

$$\begin{aligned} x^2 \equiv 1 \pmod{55} &\iff \begin{cases} x^2 \equiv 1 \pmod{5} \\ x^2 \equiv 1 \pmod{11} \end{cases} \\ &\iff \begin{cases} x \equiv \pm 1 \pmod{5} \\ x \equiv \pm 1 \pmod{11} \end{cases} \end{aligned}$$

This gives us 4 cases. Working through them,

1. $\begin{cases} x \equiv 1 \pmod{5} \\ x \equiv 1 \pmod{11} \end{cases} \implies x \equiv 1 \pmod{55}$
2. $\begin{cases} x \equiv 1 \pmod{5} \\ x \equiv -1 \pmod{11} \end{cases} \implies x \equiv 21 \pmod{55}$
3. $\begin{cases} x \equiv -1 \pmod{5} \\ x \equiv 1 \pmod{11} \end{cases} \implies x \equiv 34 \pmod{55}$
4. $\begin{cases} x \equiv -1 \pmod{5} \\ x \equiv -1 \pmod{11} \end{cases} \implies x \equiv 54 \pmod{55}$

Thus we have the four solutions $x \equiv 1 \pmod{55}$, $x \equiv 21 \pmod{55}$, $x \equiv 34 \pmod{55}$, and $x \equiv 54 \pmod{55}$.

Note: Generally, if $m = p_1^{r_1} p_2^{r_2} \cdots p_k^{r_k}$, then $x^2 \equiv 1 \pmod{m}$ is equivalent to the system of congruences

$$\begin{aligned} x^2 &\equiv 1 \pmod{p_1^{r_1}} \\ x^2 &\equiv 1 \pmod{p_2^{r_2}} \\ &\vdots \\ x^2 &\equiv 1 \pmod{p_k^{r_k}} \end{aligned}$$

Also, generally, $a \equiv b \pmod{m}$ is equivalent to the system of congruences

$$\begin{aligned} a &\equiv b \pmod{p_1^{r_1}} \\ a &\equiv b \pmod{p_2^{r_2}} \\ &\vdots \\ a &\equiv b \pmod{p_k^{r_k}} \end{aligned}$$

26.1 8.1 - Fermat's Theorem

Theorem (8.1): (Fermat's Theorem): Let p be a prime. Then for every integer b ,

$$b^p \equiv b \pmod{p}$$

Additionally if $b \not\equiv 0 \pmod{p}$, then

$$b^{p-1} \equiv 1 \pmod{p}$$

Note: the parts of Fermat's theorem imply each other, since if $b \not\equiv 0 \pmod{p}$, then $b^p \equiv b \pmod{p}$ implies $b^{p-1} \equiv 1 \pmod{p}$ by multiplying by the inverse of b modulo p . Additionally it goes the other way around because if $b \not\equiv 0 \pmod{p}$, then $b^{p-1} \equiv 1 \pmod{p}$ implies $b^p \equiv b \pmod{p}$ by multiplying by b .

Note: Consider $b = 2$. Then,

$$\begin{aligned} 2^2 &\equiv 4 \equiv 2 \pmod{2} \\ 2^3 &\equiv 8 \equiv 2 \pmod{3} \\ 2^4 &\equiv 16 \equiv 0 \not\equiv 2 \pmod{4} \\ 2^5 &\equiv 32 \equiv 2 \pmod{5} \end{aligned}$$

Example: Simplify $3^{24} \pmod{7}$. Since $3^6 \equiv 1 \pmod{7}$ by Fermat's theorem, we have $3^{24} = (3^6)^4 \equiv 1^4 \equiv 1 \pmod{7}$.

Example: Simplify $7^{182} \pmod{19}$. Fermat says that $7^{18} \equiv 1 \pmod{19}$, so $7^{182} = 7^{18 \cdot 10 + 2} = (7^{18})^{10} \cdot 7^2 \equiv 1^{10} \cdot 49 \equiv 11 \pmod{19}$.

Corollary (8.2): Let p be prime and $b \not\equiv 0 \pmod{p}$. If $x \equiv y \pmod{p-1}$ then $b^x \equiv b^y \pmod{p}$.

Proof: Suppose $x \equiv y \pmod{p-1}$, so that

$$x = y + (p-1)k \text{ for some integer } k$$

Then $b^x = b^{y+(p-1)k} = (b^{p-1})^k b^y \equiv 1^k b^y \equiv b^y \pmod{p}$. □

So when working modulo p , we can reduce exponents modulo $p-1$.

Example: Simplify $\pmod{5}$.

$$\begin{aligned} 127^{53} + 58^{70} &\equiv 2^{53} + 3^{70} \pmod{5} \\ &\equiv 2^1 + 3^2 \pmod{5} \\ &\equiv 2 + 9 \equiv 1 \pmod{5} \end{aligned}$$

Note: Fermat's Theorem says that if p is prime and $b \not\equiv 0 \pmod{p}$, then

$$b^{p-1} \equiv 1 \pmod{p}$$

So if n is a positive integer and we find a $b \not\equiv 0 \pmod{p}$ such that $b^{n-1} \not\equiv 1 \pmod{p}$, then n is not prime. For instance, $2^4 = 16 \equiv 1 \pmod{5}$, so this does not show that 5 is not prime. However, $2^4 = 16 \equiv 0 \not\equiv 1 \pmod{4}$, so this shows that 4 is not prime.

Also, if $2^{n-1} \not\equiv 1 \pmod{n}$, then n is composite.

Example: Let's show $n = 133$ is composite by evaluating $2^{132} \pmod{133}$. We have that

$$\begin{aligned} 2^2 &\equiv 4 \pmod{133} \\ 2^4 &\equiv 16 \pmod{133} \\ 2^8 &\equiv 256 \equiv 123 \pmod{133} \\ 2^{16} &\equiv 123^2 \equiv 100 \pmod{133} \\ &\vdots \\ 2^{128} &\equiv 4 \pmod{133} \end{aligned}$$

Thus, $2^{132} = 2^{128} \cdot 2^4 \equiv 4 \cdot 16 \equiv 64 \not\equiv 1 \pmod{133}$, so 133 is composite.

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Exam

28 Lecture 27

Lemma (8.4): Let p be a prime and $b \not\equiv 0 \pmod{p}$. Then the set of numbers

$$\{b, 2b, 3b, \dots, (p-1)b\} \pmod{p} = \{1, 2, 3, \dots, p-1\} \pmod{p}$$

Example: If we have $p = 11, b = 3$, then

$$\begin{aligned} \{b, 2b, 3b, \dots, 10b\} &= \{3, 6, 9, 1, 4, 7, 10, 2, 5, 8\} \\ &= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \end{aligned}$$

Proof: Note that if $1 \leq k \leq p-1$ then $kb \not\equiv 0 \pmod{p}$.

To see the elements of $\{b, 2b, \dots, (p-1)b\}$ are different, suppose $ib \equiv jb \pmod{p}$ for some $1 \leq i, j \leq p-1$. Then, multiply by b^{-1} to get $i \equiv j \pmod{p}$. Since $1 \leq i, j \leq p-1$, this implies $i = j$. Thus, the elements of $\{b, 2b, \dots, (p-1)b\}$ are different. Since there are $p-1$ elements in $\{b, 2b, \dots, (p-1)b\}$ and $p-1$ elements in $\{1, 2, \dots, p-1\}$, the two sets must be equal. $\square$

Theorem (Fermat's (Little) Theorem): Let p be a prime. Then,

1. For any integer b

$$b^p \equiv b \pmod{p}$$

2. If $b \not\equiv 0 \pmod{p}$ then

$$b^{p-1} \equiv 1 \pmod{p}$$

Proof: We know that these are equivalent to each other. Then, we will prove b.

Let $b \not\equiv 0 \pmod{p}$. Then by lemma

$$\begin{aligned} \{b, 2b, 3b, \dots, (p-1)b\} &= \{1, 2, 3, \dots, p-1\} \\ \implies b \cdot 2b \cdot 3b \cdots (p-1)b &= 1 \cdot 2 \cdot 3 \cdots (p-1) \\ b^{p-1} \cdot (p-1)! &= (p-1)! \end{aligned}$$

Since $\gcd((p-1)!, p) = 1$, we can divide both sides by $(p-1)!$ to get $b^{p-1} \equiv 1 \pmod{p}$. $\square$

Let $\phi(n)$ denote the number of elements in the set $T = \{1 \leq k \leq n : \gcd(k, n) = 1\}$. For instance, $\phi(10) = 4$ since the elements of T are 1,3,7,9.

Then

$$\begin{aligned} \phi(p) &= p-1 \text{ for prime } p \\ \phi(2) &= 1 \\ \phi(3) &= 2 \\ \phi(5) &= 4 \\ \phi(7) &= 6 \end{aligned}$$

for primes. But $\phi(4) = 2$

28.1 8.9 - Euler's Theorem

Fact: The product of two or more invertibles $(\text{mod } n)$ is also invertible. Additionally, the inverse of ab is $(ab)^{-1} = b^{-1}a^{-1}$.

Suppose $\gcd(b, n) = 1$ so that b is invertible. Let $bT = \{ba \pmod{n} : a \in T\}$. Then, bT is the set of invertibles $(\text{mod } n)$. Thus, $|bT| = |T| = \phi(n)$.

Theorem: $bT = T$.

Proof: Everything in bT is a product of invertibles, and thus is invertible, and $bT \subseteq T$.

Then, given any $a \in T$, we can write

$$a \equiv b(b^{-1}a) \pmod{n}$$

Since $b^{-1}a \in T$, we have $b^{-1}a \equiv bT$ so $a \equiv b(b^{-1}a) \equiv bT$. Thus, $T \subseteq bT$. □

Theorem (8.9): (Euler's Theorem): Suppose n positive and $\gcd(b, n) = 1$, then

$$b^{\phi(n)} \equiv 1 \pmod{n}$$

Proof: By the lemma, $T = bT$. Thus,

$$\begin{aligned} \prod_{a \in T} a &\equiv \prod_{x \in bT} x \\ &\equiv \prod_{a \in T} ba \pmod{n} \\ \left(\prod_{a \in T} a\right) &\equiv b^{\phi(n)} \left(\prod_{a \in T} a\right) \pmod{n} \end{aligned}$$

Since $\prod_{a \in T} a$ is a product of invertibles, it is invertible mod n . Multiply by $(\prod_{a \in T} a)^{-1}$ to get $1 \equiv b^{\phi(n)} \pmod{n}$. □

Remark: If $n = p$ a prime, then $\phi(p) = p - 1$ and this is Fermat's

Example: $n = 14$ has $\phi(14) = 6$ since the elements of T are 1,3,5,9,11,13. Then, if $\gcd(b, 14) = 1$, then $b^6 \equiv 1 \pmod{14}$.

Example: Reduce $5^{362} \pmod{14}$. Then,

$$\begin{aligned} 5^{362} &\equiv 5^{6 \cdot 60 + 2} \pmod{14} \\ &\equiv (5^6)^{60} \cdot 5^2 \pmod{14} \\ &\equiv 1^{60} \cdot 25 \pmod{14} \\ &\equiv 11 \pmod{14} \end{aligned}$$

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Recall: Euler's theorem says that if n is a positive integer and $\gcd(b, n) = 1$, then $b^{\phi(n)} \equiv 1 \pmod{n}$, where $\phi(n) = |\{1 \leq k \leq n : \gcd(k, n) = 1\}|$.

Corollary (8.1): Let n be a positive integer and suppose $\gcd(b, n) = 1$. Then, if $x \equiv y \pmod{\phi(n)}$ then

$$b^x \equiv b^y \pmod{n}$$

Proof: $x \equiv y \pmod{\phi(n)}$ implies $x = y + \phi(n)k$ for some integer k . Then,

$$\begin{aligned} b^x &= b^{y+\phi(n)k} \\ &= (b^{\phi(n)})^k b^y \\ &\equiv 1^k b^y \pmod{n} \\ &\equiv b^y \pmod{n} \end{aligned}$$

□

Example: Reduce $47^{50} \pmod{15}$. This is

$$\begin{aligned} 47^{50} &\equiv 2^{50} \pmod{15} \text{ since } 47 \equiv 2 \pmod{15} \\ &\equiv 2^2 \pmod{15} \text{ since } \phi(15) = 8, 50 \pmod{8} = 2 \\ &\equiv 4 \pmod{15} \end{aligned}$$

To effectively use Euler's theorem, we need to be able to compute $\phi(n)$ efficiently.

Proposition (8.7): If p is a prime and $k \geq 1$, then

$$\phi(p^k) = p^k - p^{k-1} = p^{k-1}(p-1)$$

Proof: There are p^k integers a such that $1 \leq a \leq p^k$. Then, note that $\gcd(a, p^k) \neq 1 \iff p|a \iff a = pt$ for some integer t . There are p^{k-1} integers a such that $1 \leq a \leq p^k$ and $p|a$, since there are p^{k-1} integers t such that $1 \leq t \leq p^{k-1}$. Thus, there are $p^k - p^{k-1}$ integers a such that $1 \leq a \leq p^k$ and $\gcd(a, p^k) = 1$. □

Proposition (8.6): If $\gcd(m, n) = 1$, then $\phi(mn) = \phi(m)\phi(n)$.

Proof: Use the 2 previous propositions.

Example: $\phi(100) = \phi(2^2 \cdot 5^2) = \phi(2^2) \cdot \phi(5^2) = (2^2 - 2^1)(5^2 - 5^1) = 40$.

Example: $\phi(120) = \phi(2^3 \cdot 3 \cdot 5) = \phi(2^3) \cdot \phi(3) \cdot \phi(5) = (2^3 - 2^2)(3 - 1)(5 - 1) = 32$.

Theorem (8.8): Let $n = p_1^{a_1} p_2^{a_2} p_5^{a_5} \dots p_r^{a_r}$ be the prime factorization of n . Then,

1. $\phi(n) = \prod_{i=1}^r p_i^{a_i} - p_i^{a_i-1}$
2. $\phi(n) = n \prod_{i=1}^r (1 - \frac{1}{p_i}) = n \prod_{p|n, p \text{ prime}} (1 - \frac{1}{p})$

Proof: 1.

$$\begin{aligned} \phi(n) &= \phi(p_1^{a_1} p_2^{a_2} p_5^{a_5} \dots p_r^{a_r}) \\ &= \phi(p_1^{a_1}) \cdot \phi(p_2^{a_2}) \cdot \phi(p_5^{a_5}) \dots \phi(p_r^{a_r}) \\ &= (p_1^{a_1} - p_1^{a_1-1})(p_2^{a_2} - p_2^{a_2-1})(p_5^{a_5} - p_5^{a_5-1}) \dots (p_r^{a_r} - p_r^{a_r-1}) \quad (1) \\ &= p_1^{a_1} (1 - \frac{1}{p_1}) \cdot p_2^{a_2} (1 - \frac{1}{p_2}) \cdot p_5^{a_5} (1 - \frac{1}{p_5}) \dots p_r^{a_r} (1 - \frac{1}{p_r}) \\ &= p_1^{a_1} p_2^{a_2} p_5^{a_5} \dots p_r^{a_r} \cdot (1 - \frac{1}{p_1})(1 - \frac{1}{p_2})(1 - \frac{1}{p_5}) \dots (1 - \frac{1}{p_r}) \\ &= n \prod_{i=1}^r (1 - \frac{1}{p_i}) \quad (2) \end{aligned}$$

□

Example: The only prime divisors of 100 are 2 and 5, so $\phi(100) = 100(1 - \frac{1}{2})(1 - \frac{1}{5}) = 100 \cdot \frac{1}{2} \cdot \frac{4}{5} = 40$.

Example: The only prime divisors of 120 are 2, 3, 5, so $\phi(120) = 120(1 - \frac{1}{2})(1 - \frac{1}{3})(1 - \frac{1}{5}) = 120 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5} = 32$.

Sketch of Proof of Prop 8.6: Write $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$. Then, define the function

$$F : \mathbb{Z}_{mn} \rightarrow \mathbb{Z}_m \times \mathbb{Z}_n$$

by $F(a) = (a \bmod m, a \bmod n)$. Then, F is a bijection. Additionally, assuming $\gcd(m, n) = 1$, the CRT says that $a \equiv b \bmod mn$ if and only if $a \equiv b \bmod m$ and $a \equiv b \bmod n$. Thus, F restricts to a bijection between the invertibles $\bmod mn$ and the pairs of invertibles $\bmod m$ and $\bmod n$. Thus, $\phi(mn) = \phi(m) \cdot \phi(n)$. $\square$

Note: In fact, it's a ring isomorphism because $F(a+b) = (a+b \bmod m, a+b \bmod n) = F(a) + F(b)$ and $F(ab) = (ab \bmod m, ab \bmod n) = F(a)F(b)$.

30 Lecture 29

30.1 8.3 - Wilson's Theorem

Theorem (8.12): (Wilson's Theorem): Let p be a prime. Then,

$$(p-1)! \equiv -1 \pmod{p}$$

Example:

- $(3-1)! = 2! = 2 \equiv -1 \pmod{3}$
- $(5-1)! = 4! = 24 \equiv -1 \pmod{5}$
- $(7-1)! = 6! = 720 \equiv -1 \pmod{7}$

Proof: We know that

$$(p-1)! = 1 \cdot 2 \cdot 3 \cdots (p-1)$$

Then, every b with $1 \leq b \leq p-1$ has an inverse $\bmod p$ because $\gcd(b, p) = 1$.

Note that 1 is its own inverse, as is $p-1 \equiv -1 \pmod{p}$.

However, these are the only two elements with this property. We proved previously that the only solutions of $x^2 \equiv 1 \pmod{p}$ are $x \equiv \pm 1 \pmod{p}$. Thus, every other element b with $1 \leq b \leq p-1$ has an inverse b^{-1} such that $b^{-1} \neq b$.

So,

$$\begin{aligned} (p-1)! &\equiv 1 \cdot (p-1) \cdot \prod_{b \in S} bb^{-1} \pmod{p} \\ &\equiv 1 \cdot (p-1) \cdot 1 \pmod{p} \\ &\equiv -1 \pmod{p} \end{aligned}$$

Corollary (8.13): Let $n \geq 2$ be an integer. Then, n is prime iff.

$$(n-1)! \equiv -1 \pmod{n}$$

Proof: $\Rightarrow$ if n is prime, then $(n-1)! \equiv -1 \pmod n$ by Wilson's theorem.

$\Leftarrow$: Assume $(n-1)! \equiv -1 \pmod n$. Suppose $d|n$ and $1 \leq d < n$. We'll show $d = 1$. Since $1 \leq d \leq n-1$, $d|(n-1)!$.

Thus, $(n-1)! \equiv 0 \pmod d$. But $(n-1)! \equiv -1 \pmod n$ implies $(n-1)! \equiv -1 \pmod d$ because $d|n$. So $-1 \equiv (n-1)! \equiv 0 \pmod d$. This implies $d|(0 - (-1)) = 1$, so $d = 1$, and n is prime.

30.2 7.1 - Classical Cryptosystems

Suppose Alice and Bob want to share messages, and Bob wants to send a message to Alice. We have Eve, the eavesdropper. We want only Alice to be able to read the message. So even if Eve, the eavesdropper, hears or intercepts the message, she can't read or understand it.

Bob encrypts the plaintext into a form called ciphertext, which he sends to Alice. The ciphertext is unreadable/not understandable, so Eve can't read the plaintext even if she intercepts the ciphertext.

Then, Alice and Bob need to agree on a cryptosystem and a key. Suppose they are using a shift cipher, in which they shift each letter forward by 6 letters. Then, the encoding of the message "EULER" would be "KARKX". Alice can reverse this encryption.

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Take the alphabet. Then, a shift cipher compels a bijection from the alphabet to $\mathbb{Z}_{26}$. In a shift cipher, we choose a key $k \in \mathbb{Z}_{26}$, and we encrypt the letter a to $a + k \pmod{26}$.

Example: Suppose our shift cipher is defined by the key -10 . Then, we define

$$f : x \rightarrow x - 10 \pmod{26}$$

where the plaintext is

Plaintext	G	A	U	S	S
x	6	0	20	18	18
$x - 10 \pmod{26}$	22	16	10	8	8
Ciphertext	W	Q	K	I	I

Affine Ciphers: Here, choose some a with $\gcd(a, 26) = 1$ and some $b \in \mathbb{Z}_{26}$. Then, we encrypt the letter x to $ax + b \pmod{26}$.

Example: Define the function

$$f : x \rightarrow 3x + 2 \pmod{26}$$

Then,

Plaintext	E	U	L	E	R
x	4	20	11	4	17
$3x + 2 \pmod{26}$	14	10	9	14	1
Ciphertext	O	K	J	O	B

Then, to decrypt, Alice has to solve for x . That is, she has to solve (for the first letter)

$$\begin{aligned} 3x + 2 &\equiv 14 \pmod{26} \\ 3x &\equiv 12 \pmod{26} \\ x &\equiv 3^{-1} \cdot 12 \pmod{26} && \equiv 4 \pmod{26} \end{aligned}$$

etc. In general,

$$\begin{aligned}3x + 2 &\equiv y \pmod{26} \\3x &\equiv y - 2 \pmod{26} \\9 \cdot 3x &\equiv 9(y - 2) \pmod{26} \\x &\equiv 9(y - 2) \pmod{26} \\x &\equiv 9y - 18 \pmod{26} \\x &\equiv 9y + 8 \pmod{26}\end{aligned}$$

The decryption function is this

$$f : y \rightarrow 9y + 8 \pmod{26}$$

Note: Suppose we came up with an affine cipher, but $\gcd(a, 26) \neq 1$. Then, we can't decrypt the message because we can't solve for x in the equation $ax + b \equiv y \pmod{26}$. For instance, if the gcd was 2, then it would have 2 solutions (eg 2 different PT letters would encrypt to the same CT letter), so we wouldn't be able to decrypt the message.

Note: While there are 26 different shift ciphers (including the trivial cipher), for an affine cipher $x \rightarrow ax + b \pmod{26}$ there are $\phi(26) = 12$ choices for a and 26 choices for b , so there are $12 \cdot 26 = 312$ different affine ciphers. Thus, it would be better to work with a prime order alphabet, eg if we had 29 letters. Then, $\phi(29) = 28$, so there would be $28 \cdot 29 = 812$ different affine ciphers.

Vigenere Cipher: Apply a periodic sequence of shift/affine ciphers to your letters.

Hill Cipher: Apply a periodic sequence of affine ciphers to blocks of letters.

32 Lecture 31

32.1 9.1 - RSA

Shift/affine ciphers are forms of symmetric cryptosystems, in which the sender and receiver both possess the encryption/decryption key.

There are much more advanced symmetric cryptosystems, such as the DES (Data Encryption Standard) and AES (Advanced Encryption Standard).

Question: Can Bob and Alice send a message securely if they haven't agreed on a key in advance?

This can be done with public key cryptography, and Diffie and Hellman were the first to come up with a public key cryptosystem. This is a form of asymmetric cryptosystem, in which the sender and receiver have different keys.

Here, Alice (the receiver) has a public key and a private key. Everyone can see the public key, but only Alice can see the private key. Bob (the sender) uses Alice's public key to encrypt the message, and then Alice uses her private key to decrypt the message.

Note: RSA (Rivest-Shamir-Adleman) is a public key cryptosystem that relies on the difficulty of factoring large integers.

Our message can always be an integer. There are various ways to encode a message into numbers, the least of which is that every data in a computer is a number.

For Alice to be able to receive messages:

Definition (RSA Algorithm): Setup:

1. Alice chooses two large primes p and q , and computes $n = pq$ and $\phi(n) = (p-1)(q-1)$. Think VERY large, eg 100 digits each.
2. Alice chooses an integer e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$.
3. Alice calculates d (decryption key) with the property that $de \equiv 1 \pmod{\phi(n)}$, and
4. Publishes the pair (n, e) as her public key, while keeping $p, q, d, \phi(n)$ secret, with d as her private key.

Then, to encrypt:

1. Bob wants to send the plaintext m to Alice.
2. Bob looks up Alice's public key (n, e) , and computes the ciphertext $c \equiv m^e \pmod{n}$.
3. Bob sends c to Alice.

Then, to decrypt:

1. Alice receives c from Bob.
2. Alice computes $m \equiv c^d \pmod{n}$ and recovers the plaintext m .

Note: This can be done thanks to Euler's theorem. That is

$$\begin{aligned}
 c^d &\equiv (m^e)^d \equiv m^{ed} \pmod{n} \\
 &\equiv m^{1+k\phi(n)} \pmod{n} \text{ for some integer } k \\
 &\equiv m \cdot (m^{\phi(n)})^k \pmod{n} \\
 &\equiv m \cdot 1^k \pmod{n} \\
 &\equiv m \pmod{n}
 \end{aligned}$$

Because $ed \equiv 1 \pmod{\phi(n)}$, we can write $ed = 1 + k\phi(n)$ for some integer k . Then, we can use Euler's theorem to show that $c^d \equiv m \pmod{n}$.

Example: (Toy example:) Alice chooses $p = 17, q = 31, n = 17 \cdot 31 = 527$. Then, $\phi(n) = \phi(17 \cdot 31) = 480$. Alice chooses $e = 11$, which is coprime to 480. Then, Alice calculates d such that $11d \equiv 1 \pmod{480}$.

$$11(131) + 480(-3) = 1$$

Thus, $d = 131$. Alice's public key is $(527, 11)$ and her private key is 131.

Bob wants to send $m = 50$ to Alice. Bob thus computes

$$c = 50^{11} \pmod{527}$$

We have that

$$\begin{array}{l|l}
 50^2 & 392 \pmod{527} \\
 50^4 & 392^2 \equiv 307 \pmod{527} \\
 50^8 & 307^2 \equiv 443 \pmod{527}
 \end{array}$$

and thus

$$\begin{aligned}
 50^{11} &= 50^8 \cdot 50^2 \cdot 50^1 \\
 &\equiv 443 \cdot 392 \cdot 50 \pmod{527} \\
 &\equiv 475 \pmod{527}
 \end{aligned}$$

So Bob sends $c = 475$ to Alice. Then, Alice computes

$$c^d = 475^{131} \pmod{527}$$

We have that

$$\begin{array}{c|c} 475^2 & 69 \pmod{527} \\ \vdots & \dots \\ 475^{128} & 324 \equiv 50 \pmod{527} \end{array}$$

This ends up being

$$\begin{aligned} 475^{131} &= 475^{128} \cdot 475^2 \cdot 475^1 \\ &\equiv 324 \cdot 69 \cdot 475 \pmod{527} \\ &\equiv 50 \pmod{527} \end{aligned}$$

33 Lecture 32

RSA: Alice chooses two primes p, q and computes $n = pq$ and $\phi(n) = (p-1)(q-1)$. Alice chooses e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$. Alice calculates d such that $de \equiv 1 \pmod{\phi(n)}$. Alice publishes (n, e) as her public key, while keeping $p, q, d, \phi(n)$ secret, with d as her private key.

Then, we can encrypt $c \equiv m^e \pmod{n}$ and decrypt $m \equiv c^d \pmod{n}$.

In practice, both primes are meant to be very large (hundreds of digits each).

If Eve can factor n into p and q , then she can compute $\phi(n) = (p-1)(q-1)$, and then compute d such that $de \equiv 1 \pmod{\phi(n)}$, and thus break the encryption. Alternatively, Eve could attempt to compute $\phi(n)$ directly from n . But this turns out to be just as hard as factoring n .

Decryption works because of Euler's theorem, but using Euler's Theorem requires that $\gcd(m, n) = 1$. In practice it's ok if the gcd is not 1.

Proposition (9.1): Let $n = pq$ be a product of two distinct primes and suppose $ed \equiv 1 \pmod{(p-1)(q-1)}$. Then, for any integer m ,

$$m^{ed} \equiv m \pmod{n}$$

Proof: If $\gcd(m, n) = 1$ then this follows from Euler's theorem. Else, if $\gcd(m, n) \neq 1$, then $\gcd(m, n) = p, q$, or n .

If $\gcd(m, n) = n$, then $m \equiv 0 \pmod{n}$, and $m^{ed} \equiv 0 \equiv m \pmod{n}$.

If $\gcd(m, n) = p$, then we'll prove the result $\pmod{p}$, $\pmod{q}$ and then use CRT to conclude the result $\pmod{n}$.

Since $p|m$, $m \equiv 0 \pmod{p}$ and therefore

$$m^{ed} \equiv 0 \equiv m \pmod{p}$$

Since $ed \equiv 1 \pmod{(p-1)(q-1)}$, we can write $ed = 1 + k(p-1)(q-1)$ for some integer k . Then,

$$\begin{aligned} m^{ed} &= m^{1+k(p-1)(q-1)} \\ &\equiv m \cdot (m^{(p-1)(q-1)})^k \pmod{q} \\ &\equiv m \cdot (m^{(q-1)})^{k(p-1)} \pmod{q} \\ &\equiv m \cdot 1^{k(p-1)} \pmod{q} \\ &\equiv m \pmod{q} \end{aligned}$$

By Fermat's theorem. Notice $q \nmid m$, so $\gcd(m, q) = 1$ and we can apply Fermat's theorem. We've thus shown that $p|m^{ed} - m$ and $q|m^{ed} - m$, so $n = pq|m^{ed} - m$. Thus, $m^{ed} \equiv m \pmod{n}$.

33.1 11.1 - Orders of Elements

Let n be a positive integer and suppose $\gcd(a, n) = 1$. Then, we know $a^{\phi(n)} \equiv 1 \pmod{n}$ by Euler's theorem. But sometimes $a^m \equiv 1 \pmod{n}$ for some $m < \phi(n)$.

Definition (Order): Assuming $\gcd(a, n) = 1$, the order of $a \pmod{n}$ is the smallest positive integer k such that $a^k \equiv 1 \pmod{n}$.

Remark: Such an m exists thanks to Euler's theorem.

Example: Compute $\text{ord}_7(2)$.

$$\begin{aligned} 2^1 &\equiv 2 \pmod{7} \\ 2^2 &\equiv 4 \pmod{7} \\ 2^3 &\equiv 1 \pmod{7} \end{aligned}$$

So $\text{ord}_7(2) = 3$.

Theorem (11.1): Suppose $\gcd(a, n) = 1$ and let $m_0 = \text{ord}_n(a)$. Then, $a^m \equiv 1 \pmod{n}$ if and only if $m_0|m$.

Proof: $\Rightarrow$: Assume $a^m \equiv 1 \pmod{n}$. Then, we can write $m = qm_0 + r$ for some integers q, r with $0 \leq r < m_0$. Then,

$$\begin{aligned} 1 &\equiv a^m \\ &\equiv a^{qm_0+r} \\ &\equiv (a^{m_0})^q a^r \\ &\equiv 1^q a^r \\ &\equiv a^r \pmod{n} \end{aligned}$$

Since $0 \leq r < m_0$ and m_0 is the smallest positive integer such that $a^{m_0} \equiv 1 \pmod{n}$, we must have $r = 0$. Thus, $m = qm_0$, so $m_0|m$.

$\Leftarrow$: Assume $m_0|m$. Then, we can write $m = qm_0$ for some integer q . Then,

$$\begin{aligned} a^m &= a^{qm_0} \\ &= (a^{m_0})^q \\ &\equiv 1^q \pmod{n} \\ &\equiv 1 \pmod{n} \end{aligned}$$

□

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Corollary (11.2):

1. Let p be a prime and $a \not\equiv 0 \pmod{p}$. Then, $\text{ord}_p(a) | p - 1$.
2. Let n be a positive integer and $\gcd(a, n) = 1$. Then, $\text{ord}_n(a) | \phi(n)$.

Proof: For 2, $a^{\phi(n)} \equiv 1 \pmod{n}$ by Euler's theorem, so $\text{ord}_n(a) | \phi(n)$ by the previous theorem. For 1, we can apply 2 with $n = p$ to get $\text{ord}_p(a) | \phi(p) = p - 1$. $\square$

Examples: $\pmod{7}$, we have $\phi(7) = 6$. Then, $\text{ord}_7(a) | 6$ for all a with $\gcd(a, 7) = 1$. So the possible orders are $\overline{1, 2, 3, 6}$. We have $\text{ord}_7(2) = 3$, $\text{ord}_7(3) = 6$, $\text{ord}_7(4) = 3$, $\text{ord}_7(5) = 6$, and $\text{ord}_7(6) = 2$.

Example: $\pmod{12}$, we have $\phi(12) = 4$. Then, $\text{ord}_{12}(a) | 4$ for all a with $\gcd(a, 12) = 1$. So the possible orders are $\overline{1, 2, 4}$. We have $\text{ord}_{12}(1) = 1$, $\text{ord}_{12}(5) = 2$, $\text{ord}_{12}(7) = 2$, and $\text{ord}_{12}(11) = 2$.

Example: $p = 101$ is a prime. Then, what are the possibilities for $\text{ord}_{101}(a)$ for a with $\gcd(a, 101) = 1$? We have $\phi(101) = 100$, so the possible orders are the divisors of 100.

Example: What about $\text{ord}_{100}(a)$? Well, $\phi(100) = \phi(4)\phi(25) = \phi(2^2) \cdot \phi(5^2) = (4 - 2)(25 - 5) = 40$. So the possible orders are the divisors of 40.

Recall: A mersenne number is defined for p a prime and $M_p = 2^p - 1$. If M_p is prime, then it's called a Mersenne prime.

Proposition (11.4): Let p, q primes and $q | M_p$. Then $q \equiv 1 \pmod{p}$, $q \equiv 1 \pmod{2p}$

Proof: Suppose $q | M_p$. Then since $M_p = 2^p - 1$, we get $2^p \equiv 1 \pmod{q}$.

This implies $\text{ord}_q(2) | p$. Since p is prime, $\text{ord}_q(2) = 1$ or p . If $\text{ord}_q(2) = 1$, then $2 \equiv 1 \pmod{q}$, so $q | 1$, which is impossible. Thus, $\text{ord}_q(2) = p$. Then, by the previous corollary, $p | \phi(q) = q - 1$. Thus, $q \equiv 1 \pmod{p}$.

Example: Factor $M_{11} = 2047$. If M_{11} has a nontrivial prime divisor, then it must be one that satisfies $q \leq \sqrt{2047} \approx 45.2$, and also that $q \equiv 1 \pmod{11}$, and in fact $q \equiv 1 \pmod{22}$. The only prime that satisfies these conditions is 23, and indeed $23 | 2047$ since $2047 = 23 \cdot 89$.

EXAMPLE: Show $M_7 = 127$ is prime. We check all primes $q \leq \sqrt{127} \approx 11.3$ that satisfy $q \equiv 1 \pmod{7}$, and in fact $q \equiv 1 \pmod{14}$. No such q exists.

Recall: Fermat primes are defined for $n \geq 0$ and $F_n = 2^{2^n} + 1$. If F_n is prime, then it's called a Fermat prime.

Note: Euler proved $F_5 = 2^{2^5} + 1 = 4294967297$ is composite, and indeed $641 | F_5$ since $F_5 = 641 \cdot 6700417$. How did he do it?

Proposition (11.3): Let $n \geq 2, p$ prime dividing F_n . Then, $p \equiv 1 \pmod{2^{n+2}}$.

Note: It's easiest to prove this with $\pmod{2^{n+1}}$ instead of $\pmod{2^{n+2}}$, but the stronger result is that $p \equiv 1 \pmod{2^{n+2}}$, a result of later knowledge. Proof is omitted here.

So, if $p | F_5$, then $p \equiv 1 \pmod{2^7}$, so $p \equiv 1 \pmod{128}$. This gives us a list $129, 257, 385, 513, 641, \dots$ of candidates for p . We check these candidates and find that $641 | F_5$.

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Recall: The order of $a \pmod n$ is the smallest positive integer k such that $a^k \equiv 1 \pmod n$.

Facts:

- If $\gcd(a, n) = 1$, then $\text{ord}_n(a) \mid \phi(n)$.
- $a^m \equiv 1 \pmod n$ iff. $\text{ord}_n(a) \mid m$.

35.1 11.2 - Primitive Roots

Let p be a prime, $g \not\equiv 0 \pmod p$. Then recall $\text{ord}_p(g) \mid p-1$, so in particular $\text{ord}_p(g) \leq p-1$.

Definition (Primitive Root): We say g is a primitive root $\pmod p$ if $\text{ord}_p(g) = p-1$. (as opposed to something smaller)

Example: $\text{ord}_5(2) = 4$, so 2 is a primitive root $\pmod 5$.

Example: $\text{ord}_7(2) = 3$, so 2 is not a primitive root $\pmod 7$.

HW Problem: In general

$$b^x \equiv b^y \pmod p \iff x \equiv y \pmod{\text{ord}_p(b)}$$

We have that $b^x \equiv b^y \pmod p$ implies $b^{x-y} \equiv 1 \pmod p$, so $\text{ord}_p(b) \mid x-y$, so $x \equiv y \pmod{\text{ord}_p(b)}$.

Proposition (11.5): Let p be prime and assume $\gcd(g, p) = 1$. The following are equivalent

1. g is a primitive root $\pmod p$.
2. g is a generator of the group of invertibles $\pmod p$, i.e. all elements of the group are congruent to a power of g .

Proof: The main part of $\Rightarrow$ is to consider $1, g^1, g^2, \dots, g^{p-1}$. Since $\text{ord}_p(g) = p-1$, these are all distinct $\pmod p$. The claim is that there are no repeats in this list. On the contrary suppose $g^i \equiv g^j \pmod p$ for some $0 \leq i < j \leq p-1$. Then, $g^{j-i} \equiv 1 \pmod p$, so $\text{ord}_p(g) \mid j-i$. But $0 < j-i < p-1$, which contradicts $\text{ord}_p(g) = p-1$. Thus, there are no repeats in the list, so the list contains all the invertibles $\pmod p$.

$\Leftarrow$: If g is a generator of the group of invertibles $\pmod p$, then $g^1, g^2, \dots, g^{p-1}$ are all distinct $\pmod p$ and thus $\text{ord}_p(g) = p-1$. $\square$

Proposition (11.7): Let n be positive and $\gcd(x, n) = 1$. Then, let $m = \text{ord}_n(x)$, and let i be an integer. Then

$$\text{ord}_n(x^i) = \frac{m}{\gcd(m, i)}$$

In essence, the order of x^i is the order of x divided by the gcd of the order of x and i .

Proof: Let $k = \text{ord}_n(x^i)$ and let $d = \gcd(i, m)$. We want to prove that $k = m/d$.

Since $(x^i)^{m/d} \equiv (x^m)^{i/d} \equiv 1^{i/d} \equiv 1 \pmod n$, we have $k \mid m/d$. Thus, $k \leq m/d$.

On the other hand, $x^{ik} \equiv (x^i)^k \equiv 1 \pmod n$, so $m \mid \gcd(ik, m)$. Since $d = \gcd(i, m)$, we can write $i = da$ and $m = db$ for some integers a, b with $\gcd(a, b) = 1$. Then, $db \mid \gcd(dak, db)$, so $b \mid \gcd(ak, b)$. Since $\gcd(a, b) = 1$, we have $b \mid k$, so $k \geq b = m/d$. Thus, $k = m/d$. $\square$

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Corollary (11.8): If g is a primitive root for a prime p , then

$$\text{ord}_p(g^i) = \frac{p-1}{\gcd(p-1, i)}$$

Proposition (11.6): Let g be a primitive root for an odd prime p . Then

$$g^{\frac{p-1}{2}} \equiv -1 \pmod{p}$$

Proof: Note that -1 is the only element of order 2 $\pmod{p}$. This is because for $x^2 \equiv 1 \pmod{p}$, $x \equiv \pm 1 \pmod{p}$. Since g is a primitive root, $\text{ord}_p(g) = p-1$. Then, $\text{ord}_p(g^{(p-1)/2}) = (p-1)/\gcd(p-1, (p-1)/2) = 2$. Thus, $g^{(p-1)/2} \equiv -1 \pmod{p}$. $\square$

Example: 2 is a primitive for $p = 13$. For any i , $\text{ord}_{13}(2^i) = \frac{12}{\gcd(12, i)}$. For instance, $\text{ord}_{13}(2^3) = \frac{12}{\gcd(12, 3)} = 4$.

Example: Find all elements $\pmod{13}$ with order 3. They are

$$\begin{aligned} \text{ord}_{13}(2^i) = 3 &\iff \frac{12}{\gcd(12, i)} = 3 \\ &\iff \gcd(12, i) = 4 \\ &\iff i \equiv 4 \pmod{12} \text{ or } i \equiv 8 \pmod{12} \iff 2^4 \equiv 3 \pmod{13} \text{ or } 2^8 \equiv 9 \pmod{13} \end{aligned}$$

Example: Find all the primitive roots for $p = 13$. Then, we know it must have order $p-1 = 12$. So we want

$$\begin{aligned} 12 &= \frac{12}{\gcd(12, i)} \\ \gcd(12, i) &= 1 \\ i &\equiv 1, 5, 7, 11 \pmod{12} \\ 2^1, 2^5, 2^7, 2^{11} &\equiv 2, 6, 11, 7 \pmod{13} \end{aligned}$$

Notice that there are $\phi(13-1) = \phi(12) = 4$ primitive roots $\pmod{13}$.

Theorem: Let p be a prime. There are $\phi(p-1)$ primitive roots g for p such that $1 \leq g < p$.

Proof: We can justify this under the assumption that a primitive root exists. Let g be a primitive root $\pmod{p}$. Then, the primitive roots $\pmod{p}$ are precisely the elements of the form g^i with $\gcd(i, p-1) = 1$. The number of such i is $\phi(p-1)$. $\square$

Example: $p = 101$ is prime. How many primitive roots are there? We have $\phi(100) = \phi(2^2) \cdot \phi(5^2) = (4-2)(25-5) = 40$, so there are 40 primitive roots $\pmod{101}$.

Example: $p = 103$ is prime. How many primitive roots are there? We have $\phi(102) = \phi(2) \cdot \phi(3) \cdot \phi(17) = (2-1)(3-1)(17-1) = 32$, so there are 32 primitive roots $\pmod{103}$.

Example: We will show 2 is a primitive root for $p = 101$. The question here is basically, how do we tell if a given element is a primitive root?

We can use the fact that 2 is a primitive root $\pmod{101}$ iff. $\text{ord}_{101}(2) = 100$. Since $100 = 2^2 \cdot 5^2$, we have $\text{ord}_{101}(2) = 100$ iff. $\text{ord}_{101}(2) \neq 100/2 = 50$ and $\text{ord}_{101}(2) \neq 100/5 = 20$.

Proposition (11.11): Suppose p prime and $h \not\equiv 0 \pmod{p}$. Then, tfae:

1. h is a primitive root $\pmod{p}$.
2. For each prime q dividing $p - 1$, $h^{(p-1)/q} \not\equiv 1 \pmod{p}$.

Proof: $\Rightarrow$ is clear because

$$1 < \frac{p-1}{q} < p-1 = \text{ord}_p(h)$$

$\Leftarrow$ we will prove the contrapositive. Suppose h is not a primitive root $\pmod{p}$. We want that $h^{(p-1)/q} \equiv 1 \pmod{p}$ for some prime q dividing $p - 1$. Let m be the order of h , and suppose $m < p - 1$.

Then, we know $m|p - 1$ and so $p - 1 = mk$ for $k > 1, k \in \mathbb{Z}$. Let q be a prime dividing k . Then, $k = ql$ for some integer l , so $p - 1 = mql$. Then,

$$\begin{aligned} h^{(p-1)/q} &= h^{mql/q} \\ &= h^{ml} \\ &\equiv 1 \pmod{p} \end{aligned}$$

□

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37.1 11.5 - The Discrete Log Problem

One can show that $g = 2$ is a primitive root for the prime $p = 19$.

Thus, every number $1, 2, 3, \dots, 18$ has a congruence to 2^x for some x . For instance,

$$2^x \equiv 3 \pmod{19}$$

has a solution.

The brute force approach is simply to make a table and try all of them. It turns out eventually that $2^{13} \equiv 3 \pmod{19}$, so $x = 13$ is a solution. But this is both extraordinarily tedious, inelegant, and sucks.

Definition (Discrete Log Problem): Given a prime p , a primitive root $g \pmod{p}$, and an integer h with $h \not\equiv 0 \pmod{p}$, find the integer x such that

$$g^x \equiv h \pmod{p}$$

Note that since $\text{ord}_p(g) = p - 1$,

$$g^x \equiv g^y \pmod{p} \iff x \equiv y \pmod{p-1}$$

So the answer to a discrete log problem is unique $\pmod{p-1}$. Thus, there are only finitely many possibilities.

The discrete log problem is believed to be hard, and in fact the security of many public key cryptosystems relies on the hardness of the discrete log problem.

37.2 12.1 - Diffie - Hellman Key Exchange

Suppose Alice, Bob want to use a very secure symmetric cryptosystem (like AES) to communicate, but they haven't agreed on a key in advance.

They need to agree on a key, but they don't want to exchange the key in person. Is there a way to generate the secret key while having communications over public channels, without Eve being able to figure out the key?

In 1976, Diffie and Hellman proposed the following method:

Definition (Diffie Hellman Key Exchange):

- Alice and Bob agree on a prime p and a primitive root $g \pmod{p}$. This is public information.
- Alice chooses a secret integer a and computes $h_A \equiv g^a \pmod{p}$
- Bob chooses a secret integer b and computes $h_B \equiv g^b \pmod{p}$
- Alice sends h_A to Bob, and Bob sends h_B to Alice.
- Alice computes $k \equiv h_B^a \pmod{p}$, and Bob computes $k' \equiv h_A^b \pmod{p}$. It turns out that $k \equiv k' \pmod{p}$, so they have agreed on a key k .

$$h_B^a \equiv (g^b)^a \equiv g^{ab} \equiv (g^a)^b \equiv h_A^b \pmod{p}$$

So the public information is p, g , and the private info is a for Alice and b for Bob. The shared key is $k \equiv g^{ab} \pmod{p}$.

The most obvious line of attack is to try to find a, b .

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Exam

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39.1 13.1 - Squares and Square Roots Mod Primes

We are studying quadratic congruences, eg

$$ax^2 + bx + c \equiv 0 \pmod{p}$$

For instance, $x^2 + 3x + 5 \equiv 0 \pmod{11}$ is a quadratic congruence.

The simplest quadratic congruences have the form $x^2 \equiv a \pmod{p}$.

Some questions to ask!

- Are there solutions?
- If so, how many?

Example: The equation $x^2 \equiv 4 \pmod{5}$ clearly has $x \equiv 2 \pmod{5}$ as a solution, and $x \equiv -2 \equiv 3 \pmod{5}$ as a solution.

Example: The equation $x^2 \equiv 6 \pmod{7}$ has no integer solutions

Example: The equation $x^2 \equiv 2 \pmod{7}$ has $x \equiv 3, 4 \pmod{7}$ as solutions.

We are interested in understanding which numbers are squares mod p , where p is an odd prime, $p > 2$. This is because $x^2 \equiv a \pmod{p}$ has a solution iff. a is a square $\pmod{p}$. We only consider odd primes because $x^2 \equiv a \pmod{2}$ is trivial, since $x^2 \equiv 0$ or $1 \pmod{2}$.

Example: Counting, $\pmod{5}$ has 1, 4 as nonzero squares. $\pmod{7}$ has 1, 2, 4 as nonzero squares. $\pmod{11}$ has 1, 3, 4, 5, 9 as nonzero squares.

Proposition: Let p be an odd prime, and let $a \not\equiv 0 \pmod{p}$. Then, if $x^2 \equiv a \pmod{p}$ has a solution $x \equiv b \pmod{p}$, then $x^2 \equiv a \pmod{p}$ has exactly two solutions, $x \equiv \pm b \pmod{p}$.

Recall that for a prime p , $ab \equiv 0 \pmod{p}$ implies $a \equiv 0 \pmod{p}$ or $b \equiv 0 \pmod{p}$. This is because if $ab \equiv 0 \pmod{p}$, then $p|ab$, so $p|a$ or $p|b$, so $a \equiv 0 \pmod{p}$ or $b \equiv 0 \pmod{p}$.

Proof: Assume b is a solution of $x^2 \equiv a \pmod{p}$. Then, $(-b)^2 \equiv b^2 \equiv a \pmod{p}$, so $-b$ is also a solution. We will show they are distinct $\pmod{p}$. If $b \equiv -b \pmod{p}$, then $2b \equiv 0 \pmod{p}$, which implies either that $2 \equiv 0 \pmod{p}$, which is impossible since p is odd, or that $b \equiv 0 \pmod{p}$, which is impossible since $a \not\equiv 0 \pmod{p}$. Thus, $b \not\equiv -b \pmod{p}$, so $b, -b$ are distinct solutions.

Now, we will show that there are no other solutions. Suppose c is a solution of $x^2 \equiv a \pmod{p}$. Then, $c^2 \equiv b^2 \pmod{p}$, so $p|c^2 - b^2 = (c - b)(c + b)$. Since p is prime, $p|c - b$ or $p|c + b$. Thus, $c \equiv b \pmod{p}$ or $c \equiv -b \pmod{p}$. $\square$

Let's determine all odd primes p such that $x^2 \equiv -1 \pmod{p}$ has a solution.

Proposition: Let $p > 2$ be a prime. Then, $x^2 \equiv -1 \pmod{p}$ has a solution iff. $p \equiv 1 \pmod{4}$.

Proof: $\Rightarrow$: Assume $x^2 \equiv -1 \pmod{p}$. We claim $\text{ord}_p(x) = 4$. Since $x^2 \equiv -1 \pmod{p}$, we have $x^4 \equiv 1 \pmod{p}$, so $\text{ord}_p(x) | 4$. $x \not\equiv 1 \pmod{p}$ because if x were 1 then the product of x squared would not be 1. Since $x^2 \not\equiv 1 \pmod{p}$, we have $\text{ord}_p(x) \neq 1, 2$, so $\text{ord}_p(x) = 4$.

But $\text{ord}_p(x) | \phi(p) = p - 1$, so $4 | \phi(p) = p - 1$. Thus, $p \equiv 1 \pmod{4}$.

$\Leftarrow$ Suppose $p \equiv 1 \pmod{4}$. Then, let g be a primitive root $\pmod{p}$. Then, consider $x \equiv g^{(p-1)/4} \pmod{p}$. We have $x^2 \equiv g^{(p-1)/2} \equiv -1 \pmod{p}$ by the previous proposition (Prop 11.6). $\square$

Definition: Let p be an odd prime, and $a \not\equiv 0 \pmod{p}$. We say a is a quadratic residue $\pmod{p}$ if $x^2 \equiv a \pmod{p}$ has a solution, and we say a is a quadratic nonresidue $\pmod{p}$ if $x^2 \equiv a \pmod{p}$ has no solution. Equivalently a is a square mod p .

Definition (Legendre Symbol): Let p be an odd prime, and $a \not\equiv 0 \pmod{p}$. Define the Legendre Symbol as

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue } \pmod{p} \\ -1 & \text{if } a \text{ is a quadratic nonresidue } \pmod{p} \end{cases}$$

Roughly, "Quadratic Reciprocity" is a set of rules/laws/formulas to evaluate Legendre symbols

For instance,

Example: $\left(\frac{58}{101}\right)$ is a question of whether $x^2 \equiv 58 \pmod{101}$ has a solution.

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We are looking at the Quadratic Reciprocity theorem. The fundamental question is: Given a prime $p > 2$ and an integer $a \not\equiv 0 \pmod{p}$, does $x^2 \equiv a \pmod{p}$ have a solution?

Proposition (13.1): Let p be an odd prime and $a \not\equiv 0 \pmod{p}$. Then

$$a^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}$$

Moreover, a is a square $\pmod{p} \iff a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$

Proof: The first part: Let $b \equiv a^{(p-1)/2} \pmod{p}$. Then, $b^2 \equiv a^{p-1} \equiv 1 \pmod{p}$ by Fermat's theorem. By Corollary 6.11, $b \equiv 1 \pmod{p}$ or $b \equiv -1 \pmod{p}$.

The second part: $\Rightarrow$: Let a be a square, and $x^2 \equiv a \pmod{p}$. Note $x \not\equiv 0$ because if it were, then a would be equivalent to 0 and we have a contradiction by the premise of the theorem. Then, $a^{(p-1)/2} \equiv (x^2)^{(p-1)/2} \equiv x^{p-1} \equiv 1 \pmod{p}$ by Fermat's theorem.

$\Leftarrow$: Assume $a^{(p-1)/2} \equiv 1 \pmod{p}$. Let g be a primitive root $\pmod{p}$. Then, we can write $a \equiv g^k \pmod{p}$ for some integer k . Then,

$$\begin{aligned} 1 &\equiv a^{(p-1)/2} \\ &\equiv (g^k)^{(p-1)/2} \\ &\equiv g^{k(p-1)/2} \pmod{p} \end{aligned}$$

Since g is a primitive root, $\text{ord}_p(g) = p-1$, so $g^m \equiv 1 \pmod{p}$ iff. $p-1 \mid m$. Thus, $p-1 \mid k(p-1)/2$, so $2 \mid k$. Thus, $k = 2l$ for some integer l , so $a \equiv g^k \equiv g^{2l} \equiv (g^l)^2 \pmod{p}$, so a is a square $\pmod{p}$. $\square$

Example: Is $a = 2$ a square mod 17? That is, $2^{(17-1)/2} = 2^8 \equiv 1 \pmod{17}$, so 2 is a square $\pmod{17}$.

Given an odd prime p and $a \not\equiv 0 \pmod{p}$, we defined last time the Legendre symbol $(\frac{a}{p})$ to be 1 if a is a square $\pmod{p}$, and -1 if a is a non-square $\pmod{p}$.

Proposition (13.3): Let p be an odd prime and $a, b \not\equiv 0 \pmod{p}$. Then

1. (Euler's Criterion) $(\frac{a}{p}) \equiv a^{(p-1)/2} \pmod{p}$
2. $(\frac{ab}{p}) = (\frac{a}{p})(\frac{b}{p})$
3. If $a \equiv b \pmod{p}$, then $(\frac{a}{p}) = (\frac{b}{p})$

Proof: 1 is a restatement of the previous proposition, and 3 is clear. For 2, we have

$$\begin{aligned} (\frac{ab}{p}) &\equiv (ab)^{(p-1)/2} \pmod{p} \\ &\equiv a^{(p-1)/2} b^{(p-1)/2} \pmod{p} \\ &\equiv (\frac{a}{p})(\frac{b}{p}) \pmod{p} \end{aligned}$$

$\square$

Theorem (13.4): (Quadratic Reciprocity): Let p, q be odd primes. Then,

- 1.

$$(\frac{q}{p}) = (-1)^{\frac{(p-1)(q-1)}{4}} (\frac{p}{q})$$

That is,

$$\begin{aligned} (\frac{q}{p}) &= (\frac{p}{q}) \text{ if } p \equiv 1 \text{ or } q \equiv 1 \pmod{4} \\ (\frac{q}{p}) &= -(\frac{p}{q}) \text{ if } p \equiv q \equiv 3 \pmod{4} \end{aligned}$$

2. Let p be an odd prime. Then,

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv 3 \pmod{4} \end{cases}$$

3. let p be an odd prime. Then,

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}} = \begin{cases} 1 & \text{if } p \equiv 1 \text{ or } 7 \pmod{8} \\ -1 & \text{if } p \equiv 3 \text{ or } 5 \pmod{8} \end{cases}$$

Part 1 is the main theorem; 2 and 3 are Supplementary laws conjectured by Euler and proven by Gauss at the age of 18.

Example: Find $\left(\frac{29}{59}\right)$.

First, is $29 \equiv 1 \pmod{4}$? Yes. Thus,

$$\begin{aligned} \left(\frac{29}{59}\right) &= \left(\frac{59}{29}\right) \\ &= \left(\frac{1}{29}\right) \\ &= 1 \end{aligned}$$

Example: Find $\left(\frac{11}{43}\right)$.

First, $11 \equiv 3 \pmod{4}$, $43 \equiv 3 \pmod{4}$, so

$$\begin{aligned} \left(\frac{11}{43}\right) &= -\left(\frac{43}{11}\right) \\ &= -\left(\frac{10}{11}\right) \\ &= -\left(\frac{-1}{11}\right) \\ &= -(-1)^{(11-1)/2} \\ &= -(-1)^5 \\ &= 1 \end{aligned}$$

Alternatively we could do

$$\begin{aligned} \left(\frac{11}{43}\right) &= -\left(\frac{43}{11}\right) \\ &= -\left(\frac{10}{11}\right) \\ &= -\left(\frac{2}{11}\right)\left(\frac{5}{11}\right) \\ &= -\left(\frac{2}{11}\right)\left(\frac{11}{5}\right) \\ &= -\left(\frac{2}{11}\right)\left(\frac{1}{5}\right) \\ &= -\left(\frac{2}{11}\right) \\ &= -(-1)^{(11^2-1)/8} \\ &= -(-1)^{15} \\ &= 1 \end{aligned}$$

41 Lecture 40

Recall the theorem of Quadratic Reciprocity:

Theorem (13.4): (Quadratic Reciprocity): Let p, q be odd primes. Then,

1.

$$\left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}} \left(\frac{p}{q}\right)$$

That is,

$$\begin{aligned} \left(\frac{q}{p}\right) &= \left(\frac{p}{q}\right) \text{ if } p \equiv 1 \text{ or } q \equiv 1 \pmod{4} \\ \left(\frac{q}{p}\right) &= -\left(\frac{p}{q}\right) \text{ if } p \equiv q \equiv 3 \pmod{4} \end{aligned}$$

2. Let p be an odd prime. Then,

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv 3 \pmod{4} \end{cases}$$

3. let p be an odd prime. Then,

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}} = \begin{cases} 1 & \text{if } p \equiv 1 \text{ or } 7 \pmod{8} \\ -1 & \text{if } p \equiv 3 \text{ or } 5 \pmod{8} \end{cases}$$

Example: Is $190 \pmod{197}$ a square? Note that 197 is prime, but that 190 is not. However, $190 = 2 \cdot 5 \cdot 19$, and so

$$\begin{aligned} \left(\frac{190}{197}\right) &= \left(\frac{2 \cdot 5 \cdot 19}{197}\right) \\ &= \left(\frac{2}{197}\right) \left(\frac{5}{197}\right) \left(\frac{19}{197}\right) \end{aligned}$$

We have that $\left(\frac{2}{197}\right) = -1$, as $197 \equiv 5 \pmod{8}$. Then, by the theorem, we have

$$\begin{aligned} (-1) \left(\frac{197}{5}\right) \left(\frac{197}{19}\right) &= (-1) \left(\frac{2}{5}\right) \left(\frac{7}{19}\right) \\ &= (-1)(-1) \left(-\frac{19}{7}\right) \\ &= (-1)(-1) \left(-\frac{5}{7}\right) \\ &= (-1)(-1) \left(-\frac{7}{5}\right) \\ &= (-1)(-1) \left(-\frac{2}{5}\right) \\ &= (-1)(-1) (-(-1)) \\ &= 1 \end{aligned}$$

Example: $\left(\frac{300}{383}\right)$ where 383 is prime gives us

$$\left(\frac{2^2 \cdot 3 \cdot 5^2}{383}\right) = \left(\frac{2}{383}\right)^2 \left(\frac{3}{383}\right) \left(\frac{5}{383}\right)^2 = 1 \cdot \left(\frac{3}{383}\right) \cdot 1$$

Note that we can simplify to 1 because regardless of what the Legendre symbol turns out to be, we are squaring it, and so it will be one. Then, by Quadratic Reciprocity, we have

$$\begin{aligned}\left(\frac{3}{383}\right) &= -\left(\frac{383}{3}\right) \\ &= -\left(\frac{2}{3}\right) \\ &= -(-1) = 1\end{aligned}$$

Note: For which primes $p > 5$ is 5 a square? That is, for which p does $x^2 \equiv 5 \pmod{p}$ have a solution? We want that $1 = \left(\frac{5}{p}\right)$, or $1 = \left(\frac{p}{5}\right)$. So, $p \equiv 1, 2, 3, 4 \pmod{5}$ are the possibilities. If we check, we find that only 1, 4 are equivalent to 1 mod 5, and so $x^2 \equiv 5 \pmod{p}$ has a solution iff. $p \equiv 1, 4 \pmod{5}$.

41.1 11.3 - Quadratic Congruences

SO far, we've been studying quadratic congruence of the form $x^2 \equiv a \pmod{p}$. What about quadratic congruences of the form $ax^2 + bx + c \equiv 0 \pmod{p}$? We'll assume that p is an odd prime, and that a, b, c integers. Then, the quadratic formula actually still kind of works, once suitably interpreted.

We have that

$$x \equiv (-b \pm \sqrt{b^2 - 4ac})(2a)^{-1} \pmod{p}$$

But the mod p square root may or may not exist, and we also need to find the multiplicative inverse of $2a$ mod p . Thus, we have this:

Theorem: If the Legendre Symbol $\left(\frac{b^2 - 4ac}{p}\right) = -1$, then the quadratic equation has no solutions, and if its 1, then the solutions are given by the quadratic formula.

Example: Solve $x^2 + x + 1 = 0 \pmod{13}$. Then, $b^2 - 4ac = -3$, and $\left(\frac{-3}{13}\right) = \left(\frac{-1}{13}\right)\left(\frac{3}{13}\right) = \left(\frac{13}{3}\right) = \left(\frac{1}{3}\right) = 1$ and so solutions exist. Then, finding the actual solutions, we need to find the square root of $-3 \equiv 10 \pmod{13}$, and the inverse.

By the power of bullshit, the square root is 6. Then, $x \equiv (-1 \pm \sqrt{-3})(2(1))^{-1} \pmod{13}$ which gives $x \equiv (-1 \pm 6)(7) \pmod{13}$ which is 9 and 3 mod 13.